Multichannel SAR Moving-Target Detection Based on HPD Manifold in Heterogeneous Clutter

Chenyang Luo, Fatong Zhang, Yaowen Fu, Wenpeng Zhang, Wei Yang, and Ruofeng Yu

***Abstract*— Heterogeneous clutter suppression is an important challenge for multichannel synthetic aperture radar (SAR) moving-target detection. The knowledge-aided space–time adaptive processing (STAP) method is widely used due to its superior clutter suppression performance. In this process, it is necessary to use independent and identically distributed (IID) training data to estimate the clutter covariance matrix (CCM) of the cell under test (CUT). However, heterogeneous environments such as clutter power fluctuations and moving-target contamination in training samples will significantly reduce the estimation accuracy of the CCM, which is not conducive to subsequent clutter suppression processing. To solve this problem, this article introduces the covariance matrix estimation method based on the geometric mean of the Hermitian positive-definite (HPD) manifold into the multichannel SAR moving-target detection. First, the airborne multichannel SAR echo is characterized on the HPD manifold, which better reveals the nonlinear geometric structure of the multichannel SAR echo data than in Euclidean space. Based on the echo characterization on the HPD manifold, the CCM estimation problem is transformed into the geometric mean estimation problem on the HPD manifold. At the same time, the generalized inner product (GIP) criterion is used in training sample selection to reduce the negative impact of contaminated samples. Finally, the CCM estimation based on geometric mean is applied to the knowledge-aided SAR-STAP, which effectively improves the clutter suppression performance. Given the large volume of multichannel SAR echo data, this article comprehensively evaluates algorithm performance and computational complexity and selects the geometric metric criterion for multichannel SAR moving-target detection. Based on this criterion, a multichannel SAR moving-target detection framework driven by the HPD manifold is proposed. The detection performance of the proposed framework is verified on simulated and measured clutter backgrounds.**

***Index Terms*— Clutter suppression, Hermitian positive-definite (HPD) manifold, moving-target detection, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) ground moving-target indication (GMTI) system can provide high-precision information about both the scene and moving targets. Consequently, the SAR-GMTI system is widely used in applications such as traffic control, battlefield surveillance, and other areas. However, in airborne radar systems, the movement of the platform causes clutter spectrum expansion, making it difficult to detect slow-moving targets submerged in strong clutter [1]. Clutter suppression is a key problem in SAR-GMTI processing. Classical clutter suppression methods include the displaced phase center antenna (DPCA) method [2], [3], [4], the along-track interferometry (ATI) method [5], [6], [7], and the space–time adaptive processing (STAP) method [8], [9], [10]. The DPCA and ATI methods are predominantly used in dual-channel systems. However, due to the limitation of the baseline size of the dual-channel, the range of velocity measurement and the clutter suppression ability are limited. For systems with more channels, the performance of DPCA and ATI is not optimal. STAP is the optimal processing method under the maximum signal-to-noise ratio (SNR) criterion. In SAR-STAP processing, the SAR-GMTI system uses multiple receiving channels along the track direction to generate SAR images of different channels in the same scene. After multichannel SAR image registration, joint processing is performed to suppress clutter, which is essentially spatial beamforming. Similar to traditional STAP, in this process, it is necessary to use several independent and identically distributed (IID) training data to estimate the clutter covariance matrix (CCM) of the cell under test (CUT) [11]. A commonly used method is the sampling matrix inversion (SMI) method. The Reed–Mallett–Brennan (RMB) criterion points out that the SMI method can obtain an acceptable performance loss when the number of IID training samples is greater than two times the degree of freedom (DOF) of the system [12]. However, the real-world detection scenarios are usually heterogeneous, including fluctuations in terrain, masking effects, forests, roads, and artificial objects such as bridges and buildings. These features lead to significant variations in clutter power across different range cells, making it difficult for the echo data of training samples to satisfy the IID condition. Additionally, the training samples may be contaminated by moving targets, that is, the training samples contain moving targets. Using heterogeneous samples, such as clutter power fluctuation or moving-target contamination in training samples for CCM estimation, will bring different degrees of performance loss [13]. If the power of the training samples is heterogeneous, it is difficult to obtain enough training samples with consistent statistical characteristics, so that the difference between the estimated CCM of the CUT and the real

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covariance matrix will increase, which will significantly reduce the clutter suppression performance. If the training samples contain moving targets, it will lead to target cancellation, seriously affecting the performance of moving-target detection.

To combat the heterogeneous clutter environment, many useful studies have been conducted [14], [15], [16]. For the heterogeneous scenes of clutter power fluctuations, one common approach is to design a suitable strategy to eliminate heterogeneous training samples. For instance, power selection training (PST) [17] method selects the clutter samples with the strongest power to form a deep notch in the clutter region, effectively improving the performance of clutter suppression. The power and phase selection training (PPST) [18] method further uses the phase information for training sample selection on the basis of PST, which further improves clutter suppression performance. Additionally, methods such as loaded sample matrix inversion (LSMI) [19], the direct data domain least-squares (D3LS) [20] algorithm, and knowledge-aided STAP (KA-STAP) [14] have been proposed to handle the problem of heterogeneous clutter power. The LSMI method improves the robustness of CCM estimation by applying the diagonal loading technique. The D3LS method only uses the data of the CUT for processing. Compared with the statistical STAP method, the D3LS method has significant advantages in extremely heterogeneous power scenarios, but there is a problem of aperture loss. By introducing a prior structure of CCM to modify the SMI estimator, KA-STAP achieves better clutter suppression performance than the SMI method in heterogeneous power scenarios. For the case where the training samples contain moving targets, the traditional solution is to use a nonhomogeneous detector (NHD) to eliminate the samples containing moving targets. At present, two test criteria are mainly used to eliminate training samples in engineering, namely the generalized inner product (GIP) criterion [21], [22], [23] and the adaptive power residual (APR) criterion [24], [25]. The former whitens the training data through the estimated CCM and then deletes the training samples with a large power of the whitened data. The detection statistics of the latter have the form of coherent accumulation, and the detection performance is better than the GIP criterion. However, when the training data contains strong interference targets, the estimated covariance matrix will affect the APR statistics, which can easily lead to false detection and missed detection. In the multichannel SAR-STAP scheme, the steering vector of the moving target needs to be obtained by searching, which further reduces the detection efficiency of APR-NHD. Considering the application background of multichannel SAR moving-target detection, the GIP criterion is widely used for training sample elimination. After the sample selection, target cancellation can be effectively improved. However, it does not fully consider the limited sample support problem under the condition of heterogeneous clutter power. The number of homogeneous samples available after the selection of training samples is usually limited, significantly decreasing the estimation accuracy of CCM. The accurate estimation of CCM with the support of limited samples is the key problem of clutter suppression.

In recent years, the application of matrix information geometry method in radar signal processing has attracted wide attention and has played an important role in radar target detection [26], [27], [28], [29], [30], [31], [32], recognition [33], and other fields [34]. These methods make full use of the nonlinear geometric structure of the matrix manifold and provide effective support for the nonlinear processing of radar signals. At the same time, there are also many studies on clutter suppression of airborne radar based on the manifold. Chen et al. [35] proposed a heterogeneous clutter suppression method based on Riemannian manifold affine transformation, which effectively expanded the number of homogeneous samples and improved the clutter suppression performance. However, in practical scenarios, the training data regarded as homogeneous samples may not strictly satisfy the IID condition, which will lead to the subspace spanned by the homogeneous samples may not be accurate, resulting in the decrease of the estimation accuracy of the CCM and the loss of clutter suppression performance [35]. Aubry et al. [36] and Balaji and Barbaresco [37] proposed training sample selection methods based on the geometric mean of gravity and Karcher center of gravity, respectively. When the number of training samples is limited, it can still achieve better CCM estimation accuracy and improve clutter suppression performance to a certain extent. However, the aforementioned data selection method does not fully account for computational overhead, making it unsuitable for direct application in multichannel SAR moving-target detection. The Riemann center of gravity estimation involved requires a large number of iterative operations, and the computational burden is extremely heavy. In summary, under the requirement of the multichannel SAR moving-target detection task, the existing heterogeneous clutter suppression methods are difficult to adapt to the problem of limited sample support and computational overhead at the same time. Therefore, a robust and efficient CCM estimation method with limited sample support is urgently needed to enhance the performance of heterogeneous clutter suppression and moving-target detection. This motivates the research presented in this article.

According to the foregoing discussion, we require a more effective way to combat a heterogeneous clutter environment. Therefore, a multichannel SAR moving-target detection framework based on Hermitian positive-definite (HPD) manifold is proposed in this article. First, the airborne multichannel SAR echo data is represented on the HPD manifold. Then, using the constructed HPD manifold, the CCM estimation problem is transformed into the geometric mean estimation problem on the HPD manifold. At the same time, the GIP criterion is used to select the training samples, which reduces the negative impact of the training samples contaminated by the moving target. Finally, the CCM estimation based on geometric mean is applied to the KA-STAP scheme for heterogeneous clutter suppression. Considering the problem of limited sample support and computational overhead in multichannel SAR moving-target detection, this article comprehensively considers the clutter suppression performance and computational complexity and selects the geometric metric criterion for multichannel SAR-STAP moving-target detection.

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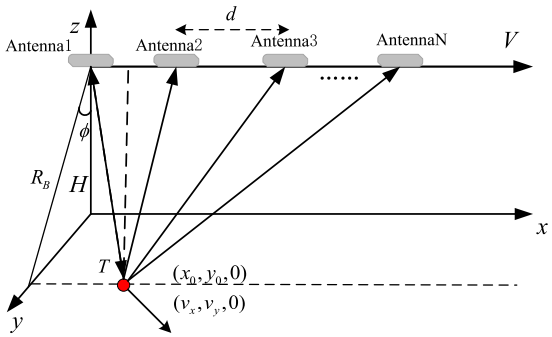


Fig. 1. Geometric model of the multichannel SAR system.

The superiority of the method described in this article is verified by experiments. The experimental results show that the method can have better heterogeneous clutter suppression performance under the condition of limited sample support.

The rest of this article is organized as follows. Section II introduces the multichannel SAR signal model and related research work. In Section III, the multichannel SAR moving-target detection framework based on the HPD manifold is introduced in detail. In Section IV, the results of clutter suppression and moving-target detection of airborne three-channel SAR under simulated and measured clutter background are given. Section V gives the conclusion of this article.

II. Airborne Multichannel SAR Signal Model and Related Work

A. Airborne Multichannel SAR Signal Model

In this article, we consider multichannel SAR systems with uniform linear arrays and side-looking, as shown in Fig. 1. The airborne radar system contains N channels distributed along the trajectory, with channel-1 transmitting and channels $1 \sim N$ receiving simultaneously. The distance between channel-n and channel-1 is $d_n (d_1 = 0)$. The uniform linear array satisfies $d_n = (n-1)d$, with d being the channel spacing. The radar system is moving at a uniform speed at altitude H with a velocity of V. ϕ is the radar down-view angle. The X- and y-axes are the direction along the trajectory (azimuthal direction) and the direction perpendicular to the trajectory (range direction), respectively. Let the moving target be located in the center of the radar beam at the moment $t_{\rm ac}$, while the position of the moving target at this moment is $(x_0, y_0, 0)$. Note the parameters of the moving target as $\boldsymbol{\vartheta}_{\rm cv} = (x_0, y_0, v_x, v_y)$, the synthetic aperture length as $L_{\rm syn}$. The synthetic aperture time is $T_{a,\rm cv} = L_{\rm syn}/(V - v_x)$.

The distance between the moving target and the nth channel can be expressed as (1), shown at the bottom of the next page. The first approximation utilizes the second-order Taylor expansion of $R_{{\rm cv},n}(t_m, \boldsymbol{\vartheta}_{\rm cv})$ at $t_m = t_{\rm ac}$, and the second approximation utilizes the truth $v_y^2 \ll (V - v_x)^2$, $v_r = v_y \sin\phi$ is the radial velocity of the moving target, and $R_B = (y_0^2 + H^2)^{1/2}$. In the SAR processing, the range compression of the target echo is performed first, and the signal of the nth channel is

$$s_{{\rm rc,cv},n}(\hat{t}, t_m, \boldsymbol{\vartheta}_{\rm cv}) = A_0 \mathrm{sinc}\left[B_r\left(\hat{t} - \frac{2R_{{\rm cv},1n}(t_m, \boldsymbol{\vartheta}_{\rm cv})}{c}\right)\right] w_a[(t_m - t_{\rm ac})] \cdot \exp\left[-j\frac{4\pi R_{{\rm cv},1n}(t_m, \boldsymbol{\vartheta}_{\rm cv})}{\lambda}\right] \quad (2)$$

where A_0 represents the scattering intensity of the moving target, λ is the radar wavelength, c is the propagation speed of electromagnetic wave in vacuum, B_r and T_r are the bandwidth and duration of the transmitting pulse, respectively, $R_{{\rm cv},1n}$ is the direct double-base distance between channel-1 and channel-n, and $w_a(\cdot)$ is the azimuthal antenna direction map, usually a rectangular window $w_a[(t_m - t_{\rm ac})] = \mathrm{rect}\left[(t_m - t_{\rm ac})/T_{a,\rm cv}\right]$. To ensure that the stationary ground clutter has consistent amplitude–phase characteristics between each channel, the channel registration process is carried out after the range compression. Since the range envelope differences of different channels can be ignored compared to the range resolution, only the phases of the clutter data need to be aligned. The registration process can be given by the following equation [38]:

$$s_{{\rm rc,cv},n}(\hat{t}, t_m, \boldsymbol{\vartheta}_c) = \mathcal{F}_{t_m}^{-1}\left\{\mathcal{F}_{t_m}\left\{s_{{\rm rc,cv},n}(\hat{t}, t_m, \boldsymbol{\vartheta}_{\rm cv}) \cdot \exp\left(j\frac{2\pi d_n^2}{4\lambda R_B}\right)\right\} \cdot \exp\left(-j2\pi f_d \frac{d_n}{2V}\right)\right\}. \quad (3)$$

Phase of the signal after channel alignment

$$\varphi(t_m, \boldsymbol{\vartheta}_{\rm cv}) \approx \left[-\frac{2\pi(V - v_x)^2}{\lambda R_B}\right](t_m - t_{\rm ac})^2 - \frac{4\pi v_r}{\lambda}(t_m - t_{\rm ac}) + \frac{2\pi d_n v_r}{\lambda V} - \frac{4\pi R_B}{\lambda}. \quad (4)$$

The aligned multichannel SAR moving-target signal can be expressed as

$$s_{rc,\rm cv}(\hat{t}, t_m, \boldsymbol{\vartheta}_{\rm cv}) = A_1 \mathrm{sinc}\left[B_r\left(\hat{t} - \frac{2R_{\rm cv}(t_m, \boldsymbol{\vartheta}_{\rm cv})}{c}\right)\right] h_{\rm cv}(t_m, \boldsymbol{\vartheta}_{\rm cv}) \Delta_{{\rm ATI,cv}}(\boldsymbol{\vartheta}_{\rm cv}) \quad (5)$$

where $A_1 = A_0 \exp(-j4\pi R_B/\lambda)$, $f_{d,\rm cv} = -2v_r/\lambda$, $K_{\rm cv} = -2(V - v_x)^2/\lambda R_B$, and $h_{\rm cv}(t_m, \boldsymbol{\vartheta}_{\rm cv})$ is azimuthal signal and its expression is

$$h_{\rm cv}(t_m, \boldsymbol{\vartheta}_{\rm cv}) = \mathrm{rect}\left(\frac{t_m - t_{\rm ac}}{T_{a,\rm cv}}\right) \exp\left\{j2\pi\left[f_{d,\rm cv}(t_m - t_{\rm ac}) + 1/2 \cdot K_{\rm cv}(t_m - t_{\rm ac})^2\right]\right\}. \quad (6)$$

$\Delta_{{\rm ATI,cv}}(\boldsymbol{\vartheta}_{\rm cv})$ is the multichannel interferometric phase vector of the moving target with the expression

$$\Delta_{ATI,\rm cv}(\boldsymbol{\vartheta}_{\rm cv}) = \left[1, \exp\left(j\frac{2\pi d_2 v_r}{\lambda V}\right), \ldots, \exp\left(j\frac{2\pi d_N v_r}{\lambda V}\right)\right]. \quad (7)$$

Subsequently, range migration correction and azimuthal compression are performed to realize SAR imaging processing. Since the moving target is noncooperative and its motion parameters are unknown, a stationary target Doppler phase history is considered to construct a matched filter in the Doppler domain for azimuthal compression, and a multichannel SAR

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image-domain signal can be obtained by utilizing the principle of stationary phase (POSP) [39]

$$s_{\mathrm{img,cv}}(\hat{t}, t_m, \boldsymbol{\vartheta}_{\mathrm{cv}}) = A_{\mathrm{cv}}\mathrm{sinc}\left[B_r\left(\hat{t} - \frac{2R_B}{c}\right)\right] \cdot \mathrm{rect}\left(\frac{t_m - t_{\mathrm{ac}} + \frac{f_{d,\mathrm{cv}}}{K_c}}{B_{\mathrm{cv}}/\Delta K}\right) \exp\left[j\pi\Delta K\left(t_m - t_{\mathrm{ac}} + \frac{f_{d,\mathrm{cv}}}{K_{\mathrm{cv}}}\right)^2\right]\Delta_{\mathrm{ATI,cv}}(\boldsymbol{\vartheta}_{\mathrm{cv}}) \quad (8)$$

where $K_c = -2V^2/\lambda R_B$ is the stationary target modulation frequency, B_{cv} is the moving-target Doppler bandwidth, and ΔK satisfies $1/\Delta K = 1/K_{\mathrm{cv}} - 1/K_c$. From (8), it can be seen that the Doppler center deviates from the zero frequency due to the effect of radial velocity of the moving target, resulting in an azimuthal shift of the signal in the SAR image domain toward the center by $\Delta t_m = -f_{d,\mathrm{cv}}/K_c = -v_r R_B/V^2$, which corresponds to an azimuthal positional shift of the target $\Delta x = -v_r R_B/V$. In addition, the azimuthal compression using the Doppler parameters of a stationary target is mismatched with the Doppler phase history of moving target, causing the imaging result to be defocused, in which case the moving-target amplitude response is the multiplication of the range direction sinc(·) and the azimuthal direction rect(·), and the width of the azimuthal magnitude response is $\Delta T = B_{\mathrm{cv}}/\Delta K$. Defocus of SAR images of moving targets reduces the target energy, and detecting moving targets in the background of stationary ground clutter is a challenging problem.

B. Heterogeneous Clutter Statistical Model

In a power-heterogeneous clutter environment, the clutter is usually modeled as a compound Gaussian random variable

$$\boldsymbol{c} = \Delta \cdot \boldsymbol{g} \quad (9)$$

where $\boldsymbol{g}$ is an N-dimensional zero-mean complex circular Gaussian random vector with $\mathrm{E}[\boldsymbol{g}\boldsymbol{g}^{\mathrm{H}}] = \Sigma$. Δ is a nonnegative random variable, and when Δ is a constant, the clutter data are homogeneous; for heterogeneous scenarios, a number of statistical models have been investigated in the literature that apply to Δ [40], [41], [42]. In this article, we use the inverse square root Gamma distribution model with the probability density function [43]

$$f_\Delta(\delta) = \frac{2(\nu-1)^\nu}{\Gamma(\nu)}\delta^{-(2\nu+1)}\exp\left(-\frac{\nu-1}{\delta^2}\right). \quad (10)$$

The power distribution of the heterogeneous clutter

$$f_T(t;\nu,n) = \frac{1}{B(n,\nu)}\left(\frac{n}{\nu-1}\right)^n \frac{t^{n-1}}{\left(1+\frac{n}{\nu-1}t\right)^{n+\nu}} \quad (11)$$

where $B(\cdot)$ denotes the Beta function, n is the number of views to be processed, and ν is the clutter texture parameter, and the larger ν is, the more homogeneous the clutter scene is. In this article, the heterogeneous clutter will be simulated using (10) and (11).

C. KA-STAP Processing Flow

In heterogeneous environments with clutter power fluctuations, it is challenging to obtain sufficient IID training samples, resulting in inaccurate CCM estimation, ultimately reducing clutter suppression performance. The knowledge-aided SAR-STAP addresses this issue by incorporating prior CCM to refine the estimator, providing an effective solution for combating such heterogeneous environments. In multichannel SAR image domain processing, let the complex-valued sequence of a certain pixel be $z_k = [z_1, z_2, \ldots, z_N]^{\mathrm{T}}$, N is the number of channels, and given the CUT $\boldsymbol{z}$(the pixel index is omitted for the convenience of derivation), the moving-target detection problem can be modeled as a binary hypothesis testing problem

$$\begin{cases} \mathcal{H}_0: & \boldsymbol{z} = \boldsymbol{c} + \boldsymbol{n} \\ \mathcal{H}_1: & \boldsymbol{z} = \boldsymbol{s} + \boldsymbol{c} + \boldsymbol{n} \end{cases} \quad (12)$$

where $\mathcal{H}_0$ and $\mathcal{H}_1$ denote the presence and absence of the target in the CUT, respectively, and $\boldsymbol{s}$, $\boldsymbol{c}$, and $\boldsymbol{n}$ denote the multichannel signal vectors corresponding to the moving target, clutter, and noise, respectively. $\boldsymbol{s} = \sigma_t \cdot \Delta_{\mathrm{ATI,cv}}(\boldsymbol{\vartheta}_{\mathrm{cv}})$. The optimal adaptive filter for STAP based on linearly constrained minimum variance (LCMV) is [44]

$$\boldsymbol{w}_{\mathrm{opt}} = \mu\boldsymbol{R}^{-1}\boldsymbol{s} \quad (13)$$

where $\mu = 1/(\boldsymbol{s}^{\mathrm{H}}\boldsymbol{R}^{-1}\boldsymbol{s})^{1/2}$, $\boldsymbol{R}$ is the CCM of CUT, which is usually obtained by using the training data that is independently and identically distributed with the CUT, and let the number of training samples used for estimation be K. The SMI estimator is

$$\hat{\boldsymbol{R}}_{\mathrm{SMI}} = \frac{1}{K}\sum_{k=1}^{K} z_k z_k^{\mathrm{H}}. \quad (14)$$

The optimal filter defined by (13) will derive the generalized likelihood ratio test (GLRT)

$$t_{\mathrm{STAP}} = \|\boldsymbol{w}_{\mathrm{opt}}^{\mathrm{H}}\boldsymbol{z}\|^2 = \frac{|\boldsymbol{s}^{\mathrm{H}}\boldsymbol{R}^{-1}\boldsymbol{z}|^2}{\boldsymbol{s}^{\mathrm{H}}\boldsymbol{R}^{-1}\boldsymbol{s}}. \quad (15)$$

$$\begin{aligned} R_{\mathrm{cv},n}(t_m, \boldsymbol{\vartheta}_{\mathrm{cv}}) &= \sqrt{[Vt_m - v_x(t_m - t_{\mathrm{ac}}) + d_n - x_0]^2 + \left[y_0 + v_y(t_m - t_{\mathrm{ac}})\right]^2 + H^2} \\ &\approx R_B + \frac{d_n^2}{2R_B} + v_r(t_m - t_{\mathrm{ac}}) + \frac{(V - v_x)d_n}{R_B}(t_m - t_{\mathrm{ac}}) + \frac{(V-v_x)^2 + v_y^2(1-\sin^2\phi)}{2R_B}(t_m - t_{\mathrm{ac}})^2 \\ &\approx R_B + \frac{d_n^2}{2R_B} + v_r(t_m - t_{\mathrm{ac}}) + \frac{(V - v_x)d_n}{R_B}(t_m - t_{\mathrm{ac}}) + \frac{(V-v_x)^2}{2R_B}(t_m - t_{\mathrm{ac}})^2 \\ &= R_B + v_r(t_m - t_{\mathrm{ac}}) + \frac{[(V - v_x)(t_m - t_{\mathrm{ac}}) + d_n]^2}{2R_B}. \end{aligned} \quad (1)$$

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Clutter suppression can be realized by processing the data vectors of each CUT in the multichannel SAR image domain by (15). However, in a heterogeneous clutter power environment, the estimated CCM deviates from the true CCM, resulting in deterioration of clutter suppression performance. To cope with this problem, literature [14] points out that the a priori CCM structure can be introduced to correct the CCM estimator using a knowledge-aided approach, which is essentially a color-loaded model. At this point, the CCM estimation based on a priori information can be expressed in the following linear, convex combinatorial form:

$$\hat{\boldsymbol{R}} = \alpha \boldsymbol{R}_0 + (1-\alpha)\hat{\boldsymbol{R}}_{\mathrm{SMI}} \tag{16}$$

where $\hat{\boldsymbol{R}}_{\mathrm{SMI}}$ is the CCM estimated from the training data, $\boldsymbol{R}_0$ is the a priori CCM, which has the priori form of $\boldsymbol{R}_0 = \sigma_c^2 1_{N\times N} + \sigma_n^2 \mathbf{I}_N$, the estimated value of the clutter power of the CUT is taken as $\hat{\sigma_c^2} = 1/N \cdot \mathbf{1}_{N\times 1}^{\mathrm{T}} z$, and the estimation of noise power σ_n^2 can be considered by selecting a sufficient number of training samples to construct the sample covariance matrix, eigen-decomposing it, and selecting the smallest eigenvalue as the estimation of noise power. The color loading coefficient α can be estimated using the MMSE criterion and its estimate is [45]

$$\alpha = \frac{\hat{\rho}}{\hat{\rho} + \|\hat{\boldsymbol{R}}_{\mathrm{SMI}} - \boldsymbol{R}_0\|^2} \tag{17}$$

$$\hat{\rho} = \frac{1}{K^2}\sum_{k=1}^{K}\|\boldsymbol{z_k}\|^4 - \frac{1}{K}\|\hat{\boldsymbol{R}}_{\mathrm{SMI}}\|^2. \tag{18}$$

Using the CCM estimator in (16) instead of the SMI estimator in (14) applied to SAR-STAP processing can effectively reduce the influence of heterogeneous samples on the CCM estimation and can significantly increase the convergence speed of the system, reduce the SCNR loss brought by the clutter power heterogeneous environment, and improve the detection performance of the moving targets.

D. Sample Selection Based on the GIP Criterion

The knowledge-aided SAR-STAP scheme well improves the performance deterioration caused by the heterogeneity of clutter power, but it does not sufficiently consider the case that the training samples contain moving targets, and when the training data used for estimating the CCM contain moving targets, the target cancellation will be generated in the filtering process, which will seriously affect the estimation performance of the CCM. Therefore, the training samples containing moving targets need to be eliminated to mitigate the effect of target cancellation. For the CUT $\boldsymbol{z_k}$, the test statistic based on the GIP criterion can be expressed as [23]

$$\eta_k = z_k^{\mathrm{H}}\hat{\boldsymbol{R}}^{-1}z_k = \left(\hat{\boldsymbol{R}}^{-\frac{1}{2}}z_k\right)^{\mathrm{H}}\left(\hat{\boldsymbol{R}}^{-\frac{1}{2}}z_k\right) = \tilde{z}_k^{\mathrm{H}}\tilde{z}_k. \tag{19}$$

From (19), the GIP value η is the energy of the sample $\boldsymbol{z_k}$ whitened by the matrix $\hat{\boldsymbol{R}}^{-1/2}$. The GIP value will deviate from the mean if the true covariance $\boldsymbol{R}_k$ of the current training sample $\boldsymbol{z_k}$ deviates from $\hat{\boldsymbol{R}}$. Accordingly, training samples containing moving targets can be filtered out.

From the above analysis, it is clear that the CCM estimation plays an important role in STAP processing. When there are enough homogeneous samples, the method based on traditional SMI estimation can obtain satisfactory performance. However, in heterogeneous environments, there is a more serious problem of missing homogeneous samples, and the number of available training samples will be further reduced after the sample selection process, and inaccurate CCM estimation will lead to a more obvious performance loss. The CCM estimation method based on the Riemannian center has achieved encouraging results under the condition of limited samples [36], [37], but the Riemannian center estimation involved requires a large number of iterative operations, the computational burden is computationally extremely burdensome. In view of the huge data load of the SAR system, it is not suitable for direct application to multichannel SAR moving-target detection. Therefore, there is an urgent need for a robust and efficient CCM estimation method to improve the performance of heterogeneous clutter suppression and moving-target detection under the restricted number of training samples.

III. Proposed Method

In this section, multichannel SAR echo data characterization on the HPD manifold is discussed, then the problem of CCM estimation is transformed into a geometric mean solving problem on the HPD manifold, and the joint GIP criterion is used for training samples screening, which is applied to realize multichannel SAR moving-target detection in a knowledge-aided SAR-STAP scheme.

A. Multichannel SAR Signal Representation Based on HPD Manifold

Assuming that there are K training samples $\{\boldsymbol{z_1}, \boldsymbol{z_1}, \ldots, \boldsymbol{z_K}\}$ for estimation of CCM, construct the sample covariance matrix $\{\hat{\boldsymbol{R}}_1, \hat{\boldsymbol{R}}_2, \ldots, \hat{\boldsymbol{R}}_K\}$ for each of the training sample data, and the CCM estimator can be regarded as a function of the matrices in the training dataset $\hat{\boldsymbol{R}} = f(\hat{\boldsymbol{R}}_1, \hat{\boldsymbol{R}}_2, \ldots, \hat{\boldsymbol{R}}_K)$. The CCM estimation problem in (14) can be regarded as the algebraic mean of the sample covariance matrix in the Euclidean space, and the estimator of (14) is equivalent to the solution of the following optimization problem:

$$\mathcal{P}_1 : \begin{cases} \hat{\boldsymbol{R}} = \arg\min\limits_{R\succ 0} \dfrac{1}{K}\displaystyle\sum_{k=1}^{K} d_E^2(\boldsymbol{R}, \hat{\boldsymbol{R}}_k) \\ \text{s.t.} \quad \hat{\boldsymbol{R}} \in \mathbb{C}^{N\times N} \end{cases} \tag{20}$$

where $\hat{\boldsymbol{R}}_k$ takes the self-covariance of the training data $\boldsymbol{z_k}\boldsymbol{z_k}^{\mathrm{H}}$, $d_E^2(\boldsymbol{R}, \hat{\boldsymbol{R}}_k) = \|\hat{\boldsymbol{R}}_k - \boldsymbol{R}\|_F^2 = tr\left[(\hat{\boldsymbol{R}}_k - \boldsymbol{R})(\hat{\boldsymbol{R}}_k - \boldsymbol{R})^{\mathrm{H}}\right]$ denotes the distance between two matrices in the Euclidean space, and $\mathbb{C}^{N\times N}$ represents all complex matrices of dimension $N \times N$. Notice that the use of Euclidean distance for the metric only takes into account the power characteristics of the data and ignores the nonlinear structure of the signal itself, the sample covariance taken at this point $\boldsymbol{z_k}\boldsymbol{z_k}^{\mathrm{H}}$ is a rank 1 matrix, while the covariance matrix should satisfy the HPD constraint. To overcome this deficiency, fully consider the nonlinear geometric

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structure of the signal, an approximate HPD matrix taken from the covariance matrix $z_k z_k^{\mathrm{H}}$ is considered, which can be solved by the following constrained optimization problem:

$$\mathcal{P}_{\text{opt}}: \begin{cases} \min\limits_{\hat{\boldsymbol{R}}_k} & \|z_k z_k^{\mathrm{H}} - \hat{\boldsymbol{R}}_k\|^2 \\ \text{s.t.} & \hat{\boldsymbol{R}}_k \succeq \boldsymbol{I}. \end{cases} \tag{21}$$

The details of solving the optimization problem defined by (21) are presented in Appendix A. The solution of the optimization problem defined by (21) is [36]

$$\hat{\boldsymbol{R}}_k = \boldsymbol{U}_k \boldsymbol{\Lambda}_k \boldsymbol{U}_k^{\mathrm{H}} \tag{22}$$

where

$$\begin{aligned} \boldsymbol{\Lambda}_k &= \operatorname{diag}([\lambda_k, 1, \ldots, 1]) \\ \lambda_k &= \max(1, \|z_k\|^2). \end{aligned} \tag{23}$$

The sample covariance matrix $\hat{\boldsymbol{R}}_k$, defined through (22) for each training sample z_k, can be expressed as the feature mapping $\hat{\boldsymbol{R}} = \varphi(z)$.

B. CCM Estimation Based on the Geometric Mean of HPD Manifold

The feature mapping $\varphi(\cdot)$ defined by (22) transforms the N-dimensional complex vector z into the $N \times N$ HPD matrix $\boldsymbol{R}$, and the set of all $N \times N$ HPD matrices constitutes the eigenspace, that is, $\mathcal{M}(N) = \{\boldsymbol{R} \in \mathcal{P}(N, \mathbb{C}), \boldsymbol{R} = \boldsymbol{R}^{\mathrm{H}}, \boldsymbol{R} \succ 0\}$. In the theory of differential geometry, the eigenspace is also known as the eigenfluid, and the HPD manifold is also an eigenfluid, the length of the curve on the eigenfluid $\mathcal{M}$ is defined as $\|\gamma\| \triangleq \int_0^1 (< \mathrm{d}\gamma/\mathrm{d}t, \mathrm{d}\gamma/\mathrm{d}t >_{\gamma(t)})^{1/2} \mathrm{d}t$, where $\mathrm{d}\gamma/\mathrm{d}t$ is the tangent vector of the curve $\gamma(t)$, $< \cdot, \cdot >_y$ is the inner product operator at the point y, and $< \mathrm{d}\gamma/\mathrm{d}t, \mathrm{d}\gamma/\mathrm{d}t >_{\gamma(t)}$ denotes the square of tangent vector at the point $\gamma(t)$. The inner product on the characteristic manifold can be defined by the metric tensor

$$< y_1, y_2 >_y \triangleq y_1^{\mathrm{T}} G(y) y_2 \tag{24}$$

where $G(\cdot)$ is the metric tensor on the characteristic manifold and $G(\cdot)$ can be regarded as a positive definite matrix. The geometric distance between two points $\boldsymbol{R}_1$ and $\boldsymbol{R}_2$ on the manifold $\mathcal{M}$ is defined as being the shortest curve length connecting the two points, which is called geodesic, that is,

$$\begin{aligned} d(\boldsymbol{R}_1, \boldsymbol{R}_2) &\triangleq \min_{\gamma \in \overline{R_1 R_2}} \|\gamma\| \\ &= \min_{\gamma \in \overline{R_1 R_2}} \int_0^1 \sqrt{\left\langle \frac{d\gamma}{\mathrm{d}t}, \frac{d\gamma}{\mathrm{d}t} \right\rangle_{\gamma(t)}} \mathrm{d}t \end{aligned} \tag{25}$$

where $\overline{\boldsymbol{R}_1 \boldsymbol{R}_2}$ is the set of curves on the manifold $\mathcal{M}$ satisfying $\gamma(0) = \boldsymbol{R}_1, \gamma(1) = \boldsymbol{R}_2$ of the set of curves. It can be verified that the curves defined by (25) satisfy the distance property (positive definiteness, symmetry, and triangular inequality).

The eigenspace constituted by the set of HPD matrices is not closed under addition and scalar products (e.g., positive definite matrices are negative definite when multiplied by a negative scalar), so it is not a linear space and has a certain nonlinear structure. Characterizing the discrepancy between two points on the manifold by the Euclidean distance does not reflect its nonlinear geometric structure, which is neither accurate nor appropriate. Therefore, it is necessary to define new measures of dissimilarity for the nonlinear structure of the HPD manifold. In statistical signal-processing problems, the metric tensor $G(\cdot)$ is typically defined as the fisher information matrix

$$g_{\mathrm{ij}}(\boldsymbol{R}) = \frac{1}{2} \operatorname{tr}\left(\boldsymbol{R}^{-1} \frac{\partial \boldsymbol{R}}{\partial y_{\boldsymbol{R}}^i} \boldsymbol{R}^{-1} \frac{\partial \boldsymbol{R}}{\partial y_{\boldsymbol{R}}^j} \right) \tag{26}$$

where $y_{\boldsymbol{R}}^i$ and $y_{\boldsymbol{R}}^j$ are the vectorized elements of the HPD matrix $\boldsymbol{R}$. Under this metric tensor definition, the inner product of matrix-form tangent vectors $\boldsymbol{R}_1$ and $\boldsymbol{R}_2$ at the point $\boldsymbol{R}$ on the manifold is given by

$$\langle \boldsymbol{R}_1, \boldsymbol{R}_2 \rangle_{\boldsymbol{R}} = \operatorname{tr}(\boldsymbol{R}^{-1} \boldsymbol{R}_1 \boldsymbol{R}^{-1} \boldsymbol{R}_2). \tag{27}$$

Under this definition of the inner product, the geodesic $\gamma(t)$ between the HPD matrices $\boldsymbol{R}_1$ and $\boldsymbol{R}_2$ for $0 \le t \le 1$ can be expressed as

$$\gamma(t) = \boldsymbol{R}_1^{\frac{1}{2}} \left(\boldsymbol{R}_1^{-\frac{1}{2}} \boldsymbol{R}_2 \boldsymbol{R}_1^{-\frac{1}{2}} \right)^t \boldsymbol{R}_1^{\frac{1}{2}}. \tag{28}$$

The Riemannian distance between the HPD matrices $\boldsymbol{R}_1$ and $\boldsymbol{R}_2$ is defined as the length of the geodesic specified in (25) and can be analytically expressed as

$$\begin{aligned} d_{\boldsymbol{R}}(\boldsymbol{R}_1, \boldsymbol{R}_2) &= \sqrt{\operatorname{tr}\left(\log^2 \left(\boldsymbol{R}_1^{-\frac{1}{2}} \boldsymbol{R}_2 \boldsymbol{R}_1^{-\frac{1}{2}} \right) \right)} \\ &= \left\| \log \left(\boldsymbol{R}_1^{-\frac{1}{2}} \boldsymbol{R}_2 \boldsymbol{R}_1^{-\frac{1}{2}} \right) \right\|_F. \end{aligned} \tag{29}$$

Due to the complexity of SAR-STAP processing, the Riemannian metric often incurs significant computational overhead. Therefore, an efficient metric is required to approximate the distance on the HPD manifold. The choice of an approximate metric directly influences the extraction and utilization of data structures, necessitating a tradeoff between performance and computational cost. In Section IV-A, we will consider commonly used approximate metric criteria, fully balancing performance and computational efficiency to select the optimal metric for multichannel SAR moving-target detection tasks.

Based on the given geometric metric, each training sample is transformed into the corresponding sample covariance matrix via (22), which corresponds to different points on the manifold. The CCM estimation problem can be converted into the estimation of the geometric mean of a set of sample points on the HPD manifold. The geometric mean of the sample covariance matrices is used as the estimator for the CCM, which is to say

$$\hat{\boldsymbol{R}} = \arg \min_{\boldsymbol{R} \succ 0} \frac{1}{K} \sum_{k=1}^{K} d^2(\boldsymbol{R}, \hat{\boldsymbol{R}}_k). \tag{30}$$

It is worth noting that if we take $G(\cdot) = I$ in (24), then the distance metric between the two matrices R_1 and R_2 in (30) becomes the Euclidean distance, which is given by $d_E(\boldsymbol{R}_1, \boldsymbol{R}_2) = (\operatorname{tr}[(\boldsymbol{R}_1 - \boldsymbol{R}_2)(\boldsymbol{R}_1 - \boldsymbol{R}_2)^{\mathrm{H}}])^{1/2} = \|\boldsymbol{R}_1 - \boldsymbol{R}_2\|_F$. In this case, the algebraic mean of the covariance estimation is derived. To illustrate this more intuitively, Fig. 2 presents

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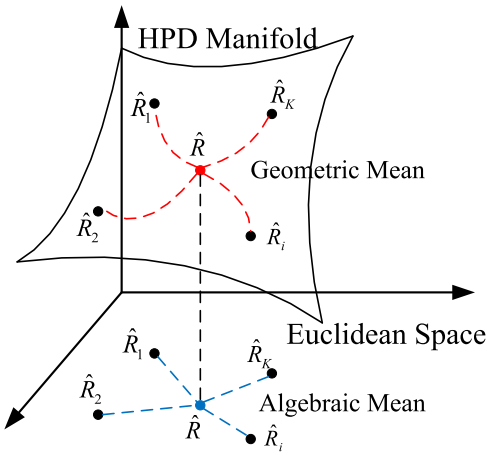


Fig. 2. Difference between the algebraic mean and the geometric mean.

the difference between the algebraic mean and the geometric mean.

From the perspective of geometric metrics, (14) is equivalent to using the algebraic mean of sample covariance matrices as an estimate of the CCM. The algebraic mean is computed in a linear space and does not take into account the nonlinear structure of the HPD manifold, resulting in a covariance estimate that may not be highly precise. Performing covariance estimation by solving for the geometric mean on the HPD manifold has the potential to achieve superior performance.

Currently, the most widely applied distance metric on the HPD manifold is the affine-invariant Riemannian metric [46] and the log-Euclidean metric [47]. This article will further compare the symmetric information divergence approximation metric of Riemannian distance [Jensen–Shannon divergence (JS divergence)]. Specifically, we consider the following three geometric metrics. The corresponding geometric mean derivation details can be found in Appendix B.

1) *Riemannian Distance:* The Riemannian distance between two covariance matrices $\boldsymbol{R}_1$ and $\boldsymbol{R}_2$ is defined as (29). By substituting (29) into (30), the corresponding geometric mean can be obtained through iteration

$$\overline{\boldsymbol{R}}_{t+1} = \overline{\boldsymbol{R}}_t^{1/2} \exp\left[\varepsilon_t \sum_{k=1}^{K} \log\left(\overline{\boldsymbol{R}}_t^{-1/2} \hat{\boldsymbol{R}}_k \overline{\boldsymbol{R}}_t^{-1/2}\right)\right] \overline{\boldsymbol{R}}_t^{1/2} \tag{31}$$

where t is the number of iterations and ε_t is the iteration step size.

2) *Log-Euclidean Distance:* The log-Euclidean distance is a concise and effective geometric metric for the HPD manifold, which fully considers the geometric characteristics of the HPD manifold. This metric maps the HPD sample matrices to the tangent space at the identity matrix via the matrix logarithm mapping. In the tangent space, the samples have a one-to-one correspondence with the samples on the original HPD manifold. Existing literature [48] has shown that the tangent space of the HPD manifold consists of symmetric matrices, which are closed under addition and scalar multiplication. This tangent space can be regarded as the minimal linear extension set of the HPD manifold, that is, the smallest linear space that contains HPD matrices. The log-Euclidean metric measures the Euclidean distance between sample matrices in the tangent space and uses it as the distance between matrices on the HPD manifold. This approach simplifies the calculations while preserving the geometric structure of the HPD manifold. Aubry et al. [36] provided the log-Euclidean distance between two covariance matrices R_1 and R_2 as follows:

$$d_L(\boldsymbol{R}_1, \boldsymbol{R}_2) = \sqrt{\mathrm{tr}\left[(\log \boldsymbol{R}_1 - \log \boldsymbol{R}_2)(\log \boldsymbol{R}_1 - \log \boldsymbol{R}_2)^{\mathrm{H}}\right]}. \tag{32}$$

Similar to the derivation of the Riemann mean, the corresponding geometric mean is obtained by substituting (32) into (30)

$$\hat{\boldsymbol{R}}_{\mathrm{LE}} = \exp\left(\frac{1}{K}\sum_{k=1}^{K} \log\ \hat{\boldsymbol{R}}_k\right). \tag{33}$$

3) *JS Divergence:* The computation based on the Riemannian distance is highly complex and will consume large computational resources in the specific implementation. The local equivalent relation between Kullback–Leibler (KL) divergence and Riemannian distance can be considered for approximate calculation [49]

$$d_{\boldsymbol{R}}^2(\boldsymbol{R}, \boldsymbol{R} + \mathrm{d}\boldsymbol{R}) \approx 2d_{\mathrm{KL}}(\boldsymbol{R}, \boldsymbol{R} + \mathrm{d}\boldsymbol{R}). \tag{34}$$

However, KL divergence only satisfies positive definiteness and does not satisfy symmetry and triangular inequality, and therefore, it is not a distance metric in the strict mathematical sense. To address the lack of symmetry in the KL divergence, the JS divergence between two covariance matrices R_1 and R_2 is defined as follows:

$$\begin{aligned} d_{\mathrm{JS}}(\boldsymbol{R}_1, \boldsymbol{R}_2) &= \frac{1}{2}\left[d_{\mathrm{KL}}\left(\boldsymbol{R}_1, \frac{\boldsymbol{R}_1 + \boldsymbol{R}_2}{2}\right) + d_{\mathrm{KL}}\left(\boldsymbol{R}_2, \frac{\boldsymbol{R}_1 + \boldsymbol{R}_2}{2}\right)\right] \\ &= \log\left|\frac{\boldsymbol{R}_1 + \boldsymbol{R}_2}{2}\right| - \log\sqrt{|\boldsymbol{R}_1 \boldsymbol{R}_2|}. \end{aligned} \tag{35}$$

The corresponding geometric mean is

$$\overline{\boldsymbol{R}}_{t+1} = \left(\frac{1}{K}\sum_{k=1}^{K}\left(\frac{\overline{\boldsymbol{R}}_t + \hat{\boldsymbol{R}}_k}{2}\right)^{-1}\right)^{-1}. \tag{36}$$

C. Multichannel SAR Moving-Target Detection Framework Based on HPD Manifold

After geometric covariance estimation, the first step is to perform training sample selection to exclude samples containing moving targets. The training sample selection can be regarded as a 2-D hypothesis testing problem

$$\begin{cases} \mathcal{H}_0: & z \in \Omega_0 \\ \mathcal{H}_1: & z \in \Omega_1 \end{cases} \tag{37}$$

where Ω_0 and Ω_1 represent the sample sets without and with moving targets, respectively. After obtaining the covariance

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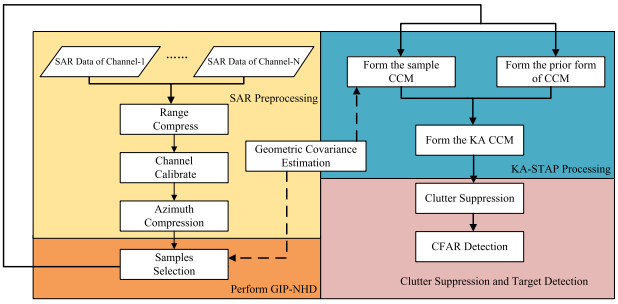


Fig. 3. Framework of multichannel SAR moving-target detection based on the HPD manifold.

estimation value through (30), it can be substituted into (19) to derive the decision criterion [23]

$$\eta_k = \boldsymbol{z}_k^{\mathrm{H}}\left(\arg\min_{\boldsymbol{R}\succ 0}\frac{1}{K}\sum_{k=1}^{K} d^2(\boldsymbol{R},\hat{\boldsymbol{R}}_k)\right)^{-1}\boldsymbol{z}_k \underset{\mathcal{H}_0}{\overset{\mathcal{H}_1}{\gtrless}} \overline{\eta} \tag{38}$$

where $\boldsymbol{d}(\cdot)$ denotes the chosen distance metric criterion and $\overline{\eta}$ represents the mean of the GIP test statistic corresponding to the training samples. After the sample selection, the remaining samples are utilized again by (30) for sample covariance estimation, which is used to replace the SMI estimator in (16). Then, (15) is employed for clutter suppression. Finally, the clutter suppression results are used for constant false alarm rate (CFAR) detection.

In summary, the proposed method mainly includes the following steps.

1) *Step 1:* The multichannel SAR image-domain signal is obtained by performing range compression, channel registration, and azimuth compression on the multichannel SAR echo signal.
2) *Step 2:* Define the sample covariance matrix $\hat{\boldsymbol{R}}_k$ for each training sample $\boldsymbol{z}$ through (22) to get the set of sample covariance matrices $\{\hat{\boldsymbol{R}}_1, \hat{\boldsymbol{R}}_2, \ldots, \hat{\boldsymbol{R}}_K\}$.
3) *Step 3:* Based on the selected geometric metric criterion, calculate the geometric mean as covariance estimates using (30).
4) *Step 4:* Select training samples using (38).
5) *Step 5:* Perform geometric covariance estimation using the remaining training samples, and the CCM estimation results are obtained by substituting into (16).
6) *Step 6:* Perform clutter suppression using (15).
7) *Step 7:* Apply the CFAR detector to the result of clutter suppression.

Fig. 3 shows the framework of multichannel SAR moving-target detection based on the HPD manifold.

IV. Experimental Results and Analysis

The performance of the proposed method is discussed in detail in this section. The simulation environment is the Windows 11 operating system, the processor is 12th Gen Intel[1] Core[2] i7-12700H, and the system memory is 32GB. The parameters of the radar system used in this experiment are given in Table I. The clutter-to-noise ratio (CNR) is set to 20 dB. Section IV-A discusses the clutter suppression performance of different methods, the selection of geometric metrics, and the detection performance of the proposed method. Section IV-A3 discusses the detection performance of the proposed method under the heterogeneous clutter environment. Sections IV-B and IV-C present the results of clutter suppression and moving-target detection under simulated and measured clutter backgrounds, respectively.

[1]Registered trademark.
[2]Trademarked.

TABLE I
Radar Parameters for Experiments

Radar Parameter	Symbol	Value
Carrier frequency	f_c	9.6GHz
Sampling frequency	f_s	120MHz
Bandwidth	B_r	100MHz
Pulse repetition frequency	PRF	1500Hz
Velocity	V	300m/s
Height	H	6km
Number of channels	N	6
Channel spacing	d	1m
Clutter to noise ratio	CNR	20dB

A. Performance Analysis

1) Performance Analysis of Clutter Suppression and Training Sample Selection: In this section, commonly used training sample selection methods in Euclidean space GIP [23] and its diagonal loading-enhanced variant, LGIP [50], are compared with the geometric mean-based GIP method. The evaluation is conducted based on the probability of moving-target samples being selected. Additionally, to assess the clutter suppression performance, we incorporate typical Euclidean space methods, including LSMI [19], PPST [18], and D3LS [20], into the comparison and evaluate their effectiveness using the output signal-to-clutter-noise ratio (SCNR). The geometric methods based on Riemannian distance, log-Euclidean distance, and JS divergence are denoted as R-GIP, LE-GIP, and JS-GIP, respectively. The clutter environment is modeled using the compound Gaussian model as defined in (9). The texture random variable is generated by the product of Gaussian clutter and the texture model described in (10), with varying heterogeneity controlled by the texture parameter ν. Moving-target signal vectors are simulated using multichannel SAR image-domain samples, as defined in (8). The experimental data are generated from 100 independent Monte Carlo experiments, with each experiment simulating 30 clutter samples and randomly injecting eight moving targets into the clutter samples. In the experiments, the signal-to-clutter ratio (SCR) and the radial velocity of the moving targets are varied, and the variations in the output SCNR and the probability of moving-target samples being selected across different methods are analyzed.

In the experiment, we set the clutter texture parameters $\nu = 20$ and $\nu = 25$ to evaluate the performance of different methods when the clutter heterogeneity changes. The diagonal loading of the LSMI method and the LGIP method is set to the estimated noise power. The PPST method first removes 20% of the training data according to the power, then removes 20% of the training data according to the phase, and uses

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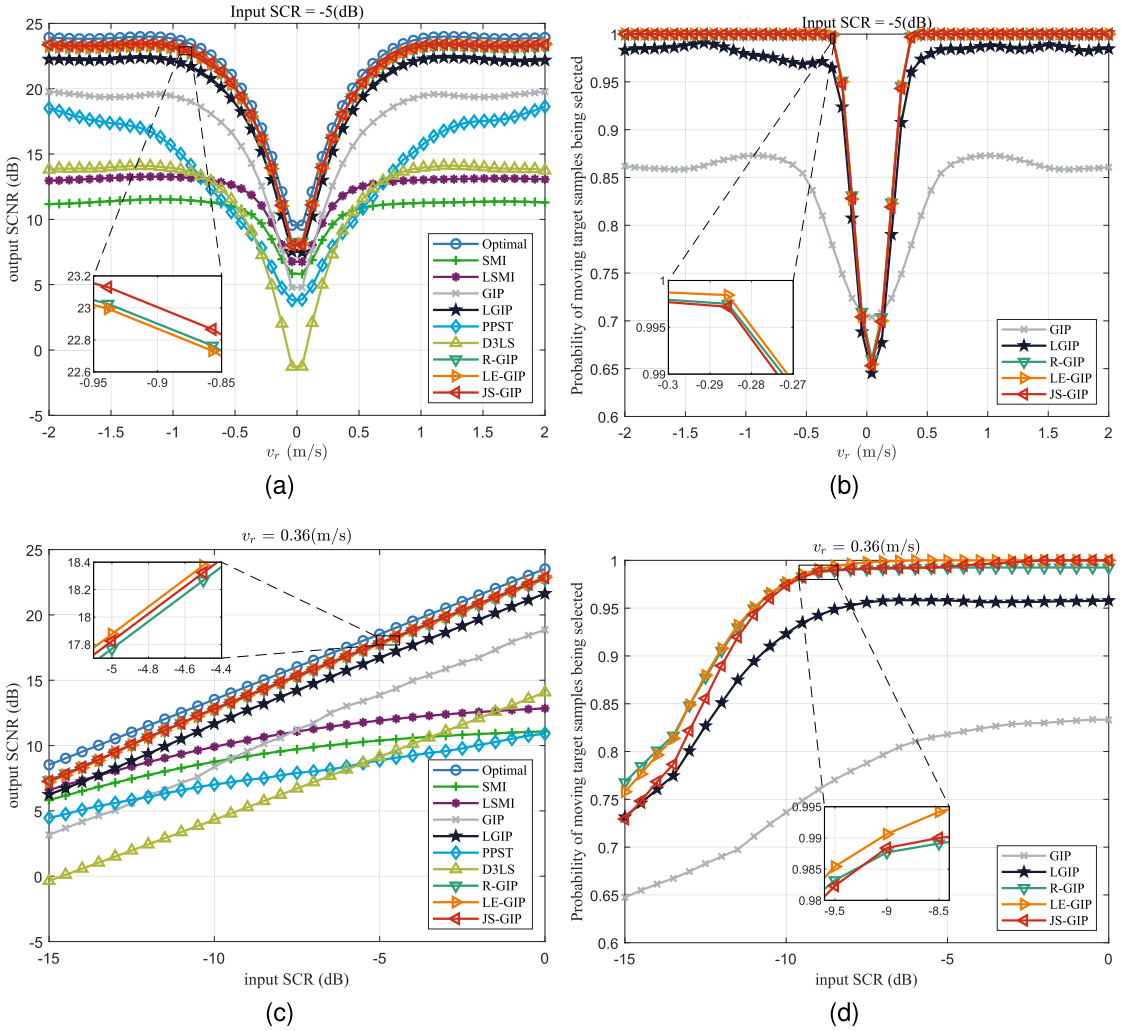


Fig. 4. Performance of clutter suppression and training sample selection ($\nu = 20$). (a) Variation of the output SCNR with radial velocity. (b) Variation of the detection of moving-target samples being selected with radial velocity. (c) Variation of the output SCNR with the input SCR. (d) Variation of detection of moving-target samples being selected with the input SCR.

the remaining training sample set to estimate the CCM. The iterative step size of the R-GIP method is $\varepsilon_t = 10^{-6}$, and the convergence condition of the R-GIP and JS-GIP methods is $\left\|\overline{\boldsymbol{R}}_{t+1} - \overline{\boldsymbol{R}}_t\right\|_F \leq 10^{-6}$. According to the engineering experience, the selection of the above iteration step size and convergence conditions can ensure the convergence of the iterative algorithm and make it converge to the Riemann mean (or JS divergence mean). First, we set SCR $= -5$ dB to evaluate the probability of moving-target samples being selected and the output SCNR for each method under varying radial velocities. Subsequently, with the moving-target radial velocity fixed at 0.36 m/s, the probability of moving-target samples being selected and the output SCNR are evaluated for each method under different input SCRs. After 100 independent Monte Carlo experiments, the results of $\nu = 20$ and $\nu = 25$ are given in Figs. 4 and 5, respectively.

The results shown in Figs. 4 and 5 indicate that the performance of all methods is improved with the increase of homogeneity. Meanwhile, methods based on geometric mean estimation on the HPD manifold have similar performance in different heterogeneity clutter environments and generally outperform the traditional methods in Euclidean space. Finally, experimental results are presented with a clutter sample size of 30, clutter texture parameter $\nu = 25$, SCR of -5 dB, and moving-target radial velocity of 0.36 m/s, where eight moving targets are randomly injected into the clutter samples. Gray boxes indicate missed detections, red boxes indicate correct detections, and black boxes indicate false alarms. The detection results for each method are shown in Fig. 6, where the x-axis represents the sample index and the y-axis is the GIP test statistic. It can be observed that the R-GIP, LE-GIP, and JS-GIP methods correctly detect samples contaminated by moving targets; the LGIP method detects all samples but has five false alarms; the GIP method has one missed detection and five false alarms. Compared with the typical methods in Euclidean space, the methods based on geometric mean estimation have a better performance.

2) Computational Complexity Analysis and Selection of Geometric Metric Criteria: According to the foregoing discussion, the methods of CCM estimation based on geometric mean have similar performance and generally outperform the traditional methods in Euclidean space. However, the computational cost of estimating the geometric mean based

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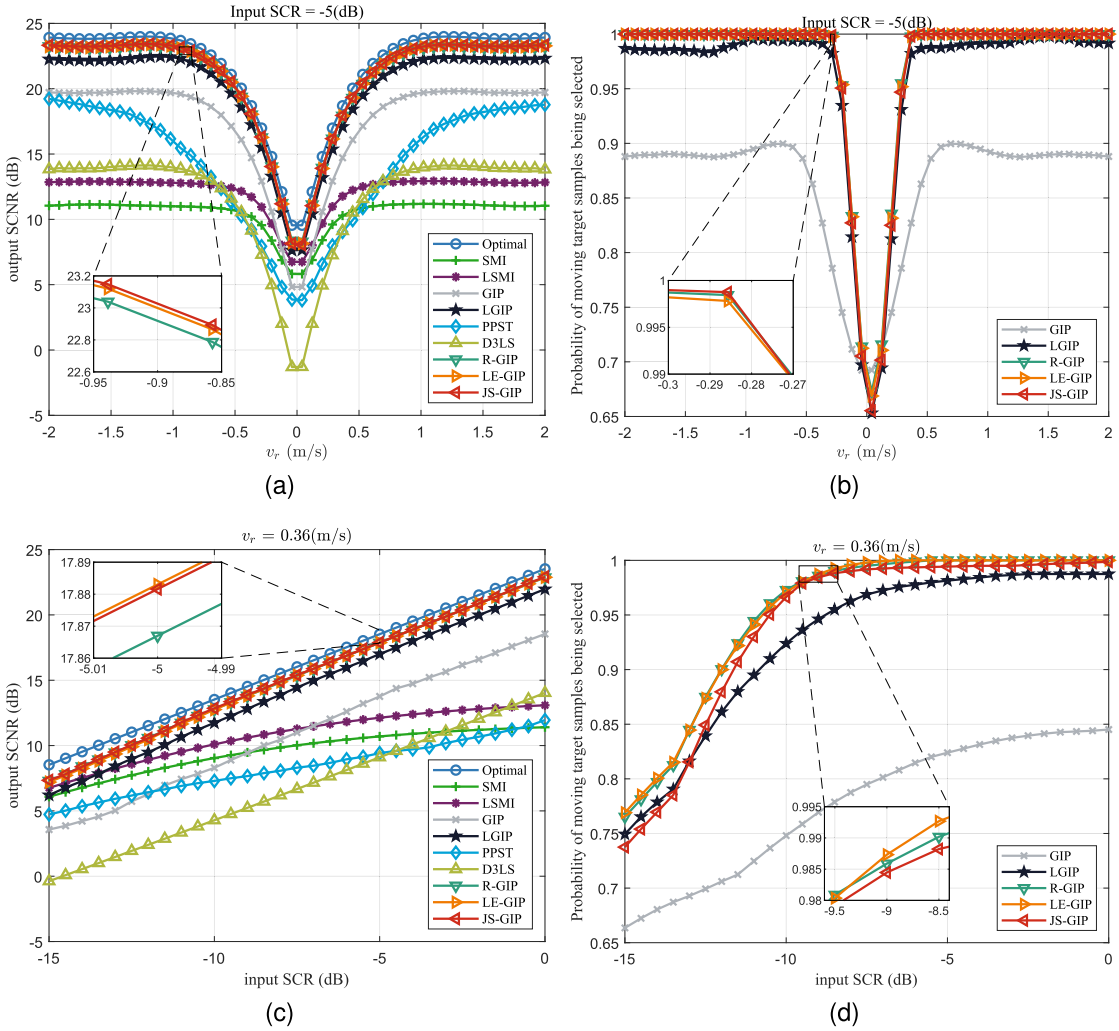


Fig. 5. Performance of clutter suppression and training sample selection ($\nu = 25$). (a) Variation of the output SCNR with radial velocity. (b) Variation of the detection of moving-target samples being selected with radial velocity. (c) Variation of the output SCNR with the input SCR. (d) Variation of detection of moving-target samples being selected with the input SCR.

on different geometric metrics is different. SAR systems usually have a heavy data load. To reduce the computational burden of SAR systems, this section comprehensively and specifically analyzes the computational complexity of different methods to estimate the CCM and then selects the geometric metric criterion suitable for multichannel SAR moving-target detection. Assuming the matrix inversion is performed via Gaussian elimination and square root operations via singular value decomposition (SVD), the computational complexities for these operations are $\mathcal{O}(N^3)$. Denote the number of channels as N, the number of training samples used to estimate the CCM as K, and the number of training samples after selection as K'. The R-GIP method and the JS-GIP method involve two iterative processes. It is assumed that the number of iterations required for convergence is N_{iter1} and N_{iter2}, respectively. For the SMI method and the LSMI method, their computational complexity is determined by the process of calculating the arithmetic mean of the covariance. The computational complexity is $\mathcal{O}(\text{KN}^2)$. The computational complexity of the PPST method depends on the computational cost of estimating the covariance from the remaining samples, which is $\mathcal{O}(K'N^2)$. The computational complexity of the GIP method and the LGIP method is determined by the calculation of the GIP test statistic, whose value is $\mathcal{O}(KN^2) + \mathcal{O}(N^3)$. For the three geometric methods, the computational cost of obtaining $\hat{\boldsymbol{R}}_k$ from a single sample $\boldsymbol{z}_k$ through feature mapping is mainly determined by eigenvalue decomposition, and the computational complexity is $\mathcal{O}(N^3)$. For K samples, the computational complexity of this step is $\mathcal{O}(\text{KN}^3)$. The flow of the three geometric methods includes two covariance estimates based on geometric means. This is also a major source of computational complexity for geometric methods. For the LE-GIP method, the computational complexity of these two steps is $\mathcal{O}(KN^3)$ and $\mathcal{O}(K'N^3)$, respectively. For the R-GIP method and the JS-GIP method, the computational complexity of these two steps is $\mathcal{O}(\text{KN}_{\text{iter1}}N^3)$ and $\mathcal{O}(K'N_{iter2}N^3)$, respectively. The computational complexity of each method can be summarized in Table II, and the computational complexity of typical methods in Euclidean space is also given in the table. Since the D3LS method is different from the statistical STAP method, there is no CCM estimation process, so it is not involved in the comparison of computational complexity. It is worth explaining that the space overhead of the geometric methods are mainly caused by the HPD matrix needed to store the

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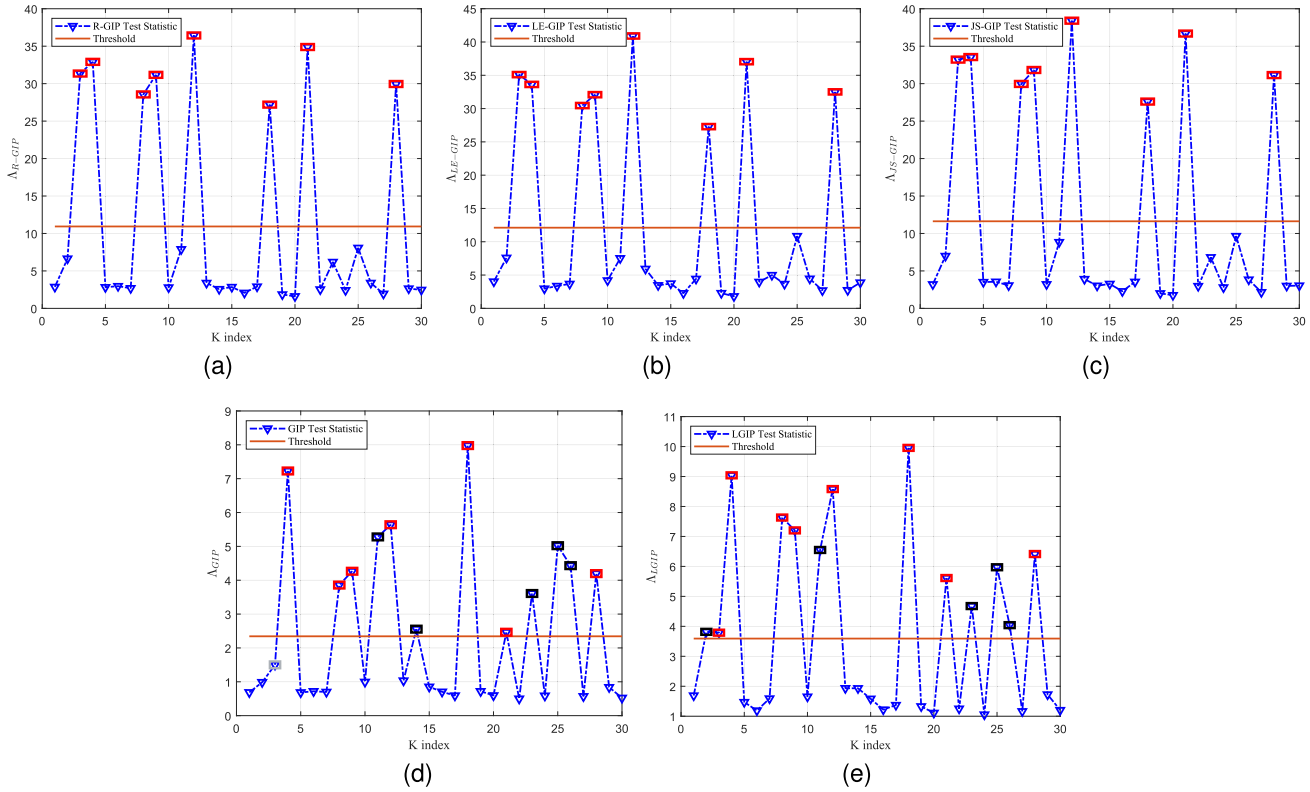


Fig. 6. Results of training sample selection. (a) R-GIP. (b) LE-GIP. (c) JS-GIP. (d) GIP. (e) LGIP.

TABLE II
COMPUTATIONAL COMPLEXITY OF DIFFERENT METHODS FOR CCM ESTIMATION

Methods	Computational complexity
R-GIP	$\mathcal{O}\left(KN_{iter1}N^3\right)+\mathcal{O}\left(K'N_{iter2}N^3\right)$
LE-GIP	$\mathcal{O}\left(KN^3\right)$
JS-GIP	$\mathcal{O}\left(KN_{iter1}N^3\right)+\mathcal{O}\left(K'N_{iter2}N^3\right)$
SMI	$\mathcal{O}\left(KN^2\right)$
LSMI	$\mathcal{O}\left(KN^2\right)$
GIP	$\mathcal{O}\left(KN^2\right)+\mathcal{O}\left(N^3\right)$
LGIP	$\mathcal{O}\left(KN^2\right)+\mathcal{O}\left(N^3\right)$
PPST	$\mathcal{O}\left(K'N^2\right)$

feature mapping defined by (22), and its space overhead is $\mathcal{O}(KN^2)$, while the space overhead of the traditional European space method is mainly caused by the storage of training samples and covariance matrices, and its space overhead is $\mathcal{O}(\mathrm{KN}+N^2)$. In the case of sufficient memory, the geometric methods can effectively improve the clutter suppression performance. As shown in Table II, considering both performance and computational complexity, the log-Euclidean distance-based geometric mean estimation for the CCM provides a good balance, ensuring high estimation accuracy with reasonable computational efficiency. Therefore, this article uses log-Euclidean distance as a geometric metric in subsequent experiments.

3) Detection Performance in Heterogeneous Clutter: In this section, the evaluation results of the detection performance for the proposed multichannel SAR moving-target detection framework based on the HPD manifold are presented. Following the analysis in Section IV-A2, we select the log-Euclidean distance as the geometric metric and combine it with KA-STAP for clutter suppression processing. Therefore, the proposed method in this article is denoted as KA-STAP (LE-NHD), while the traditional GIP method combined with KA-STAP is denoted as KA-STAP (NHD). In the experiment, 100 training samples adjacent to the CUT are used to estimate the CCM. In order to closely simulate the processing flow of a real SAR system, the experiments in this study are based on the airborne multichannel SAR imaging geometry and signal model described in Section II-A. Starting from simulating scene scattering coefficients, the entire process of SAR system operations, including echo acquisition, image registration, clutter suppression, and moving-target detection, is simulated. In the experiments, 12 moving targets were set up, which were divided into two clusters with six targets in each cluster. The azimuthal spacing between the targets is 30 m, and the radial spacing is 10 and 30 m. The radial velocity is fixed at $v_r = 1$ m/s, and the input SCR is varied to assess the change in detection probability with the input SCR. Furthermore, with the input SCR fixed at 0 dB, the radial velocity is varied to evaluate the change in detection probability with radial velocity. To assess the performance of the proposed method under different heterogeneous clutter environments, the clutter texture parameters are set to $\nu = 20$ and $\nu = 25$, respectively. Since moving targets consist of multiple pixels, the number of pixels corresponding to the main lobe width after sampling can be approximated as $Q = \gamma_{r,\mathrm{os}}\gamma_{x,\mathrm{os}}$, where $\gamma_{r,\mathrm{os}} = f_s/B_r$ and $\gamma_{x,\mathrm{os}} = PRF/B_{\mathrm{cv}}$. Since the Doppler bandwidth B_{cv} of the moving target is unknown, we use the Doppler bandwidth B_c of the stationary target to calculate $\gamma_{x,\mathrm{os}}$. By employing clustering methods, the multiple pixels of a target can be assigned to a single-target cluster. To effectively suppress false

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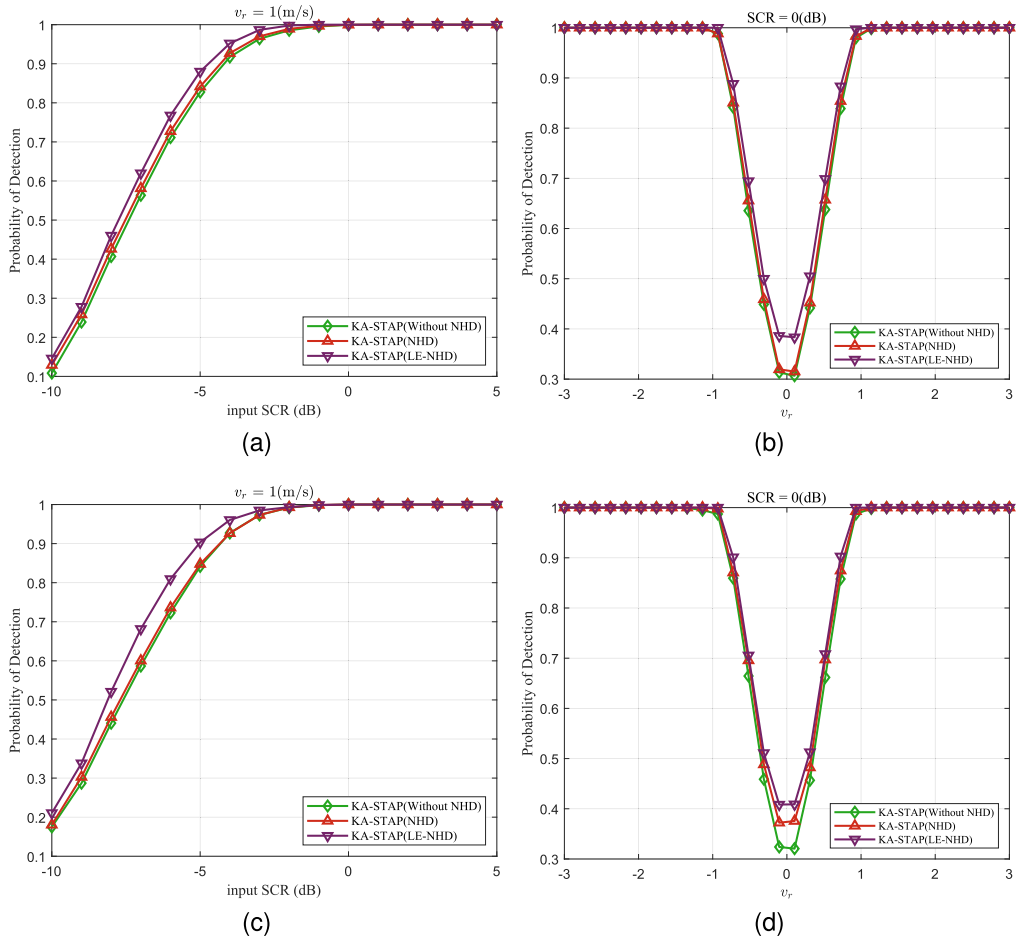


Fig. 7. Detection performance before clustering. (a) $P_d - \mathrm{SCR}(\nu = 20)$. (b) $P_d - v_r(\nu = 20)$. (c) $P_d - \mathrm{SCR}(\nu = 25)$. (d) $P_d - v_r(\nu = 25)$.

alarms, we use a clustering method that discards target clusters with fewer than $\gamma_{r,\mathrm{os}}\gamma_{x,\mathrm{os}} = 6$ pixels as false alarms. After 100 independent Monte Carlo simulations, the results before and after clustering are shown in Figs. 7 and 8, respectively.

As shown in Figs. 7 and 8, when the radial velocity is fixed, the detection probabilities of all methods gradually increase as the input SCR increases. At the same input SCR, KA-STAP (LE-NHD) achieves the best performance, while KA-STAP without NHD performs the worst. When the input SCR is fixed, the detection probabilities of all methods increase as the radial velocity increases. The KA-STAP (LE-NHD) method exhibits the narrowest notch, indicating a smaller minimum detectable velocity and the optimal detection performance. It is worth explaining that since the radial velocity is proportional to the multichannel interference phase, in general, the increase in radial velocity will lead to an increase in detection probability. But this is on the premise that the radial velocity should still be in the search space, that is, $|v_r| < \lambda V/2d$. If this range is exceeded, the interference phase along the track will be blurred, and the detection probability will decrease. In this experiment, the radial velocity search space is (−4.68,4.68 m/s), and the radial velocity change set in the experiment is within this range. It is worth noticing that, according to the results shown in Fig. 7, KA-STAP (NHD) does not exhibit a significant performance advantage over KA-STAP before clustering processing. This is because, without false alarm suppression, as long as a pixel unit detected in the moving-target region is considered a correct detection, the addition of NHD may remove some homogeneity training samples, reducing the number of training samples used to estimate the CCM. On the other hand, KA-STAP (LE-NHD), which estimates the covariance based on the geometric mean, demonstrates better robustness in CCM estimation under small training sample conditions, resulting in approximately a 5% improvement in detection probability under most conditions. The detection performance after clustering is shown in Fig. 8. It can be seen that in the results after clustering, after adding NHD for training sample selection, the contamination of training samples is effectively reduced, and the detection performance is improved to some extent. Furthermore, by estimating the CCM based on the geometric mean, the estimation accuracy of the CCM is enhanced, leading to a further increase in the detection probability. Additionally, according to the results shown in Figs. 7 and 8, with the improvement of clutter homogeneity, under the same input SCR and radial velocity, the detection probability of the same method is enhanced.

B. Detection Results in Simulated Clutter

This section presents the detection results of moving targets in simulated clutter. The radar parameters are still based on those shown in Table I. In the scenario, 12 moving targets

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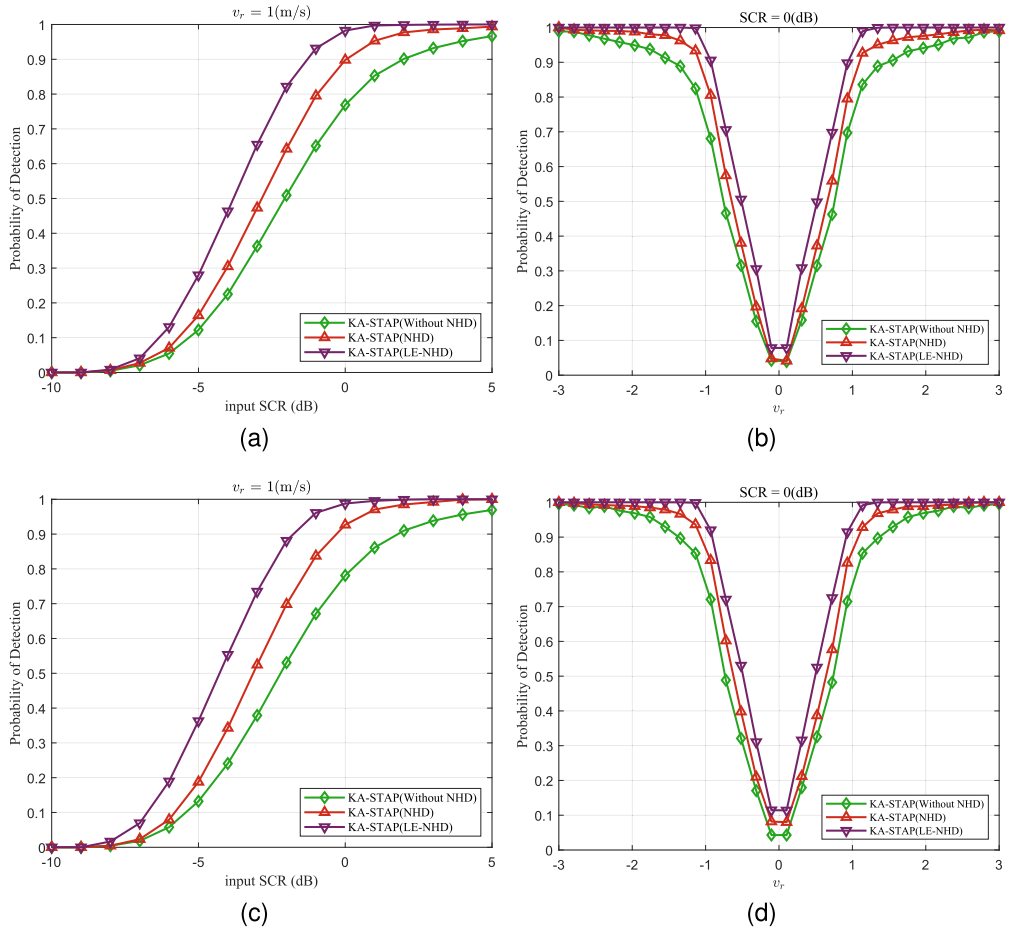


Fig. 8. Detection performance after clustering. (a) $P_d - \mathrm{SCR}(\nu = 20)$. (b) $P_d - v_r(\nu = 20)$. (c) $P_d - \mathrm{SCR}(\nu = 25)$. (d) $P_d - v_r(\nu = 25)$.

are uniformly distributed, with the target motion parameters set to $v_x = 1$ and $v_r = 1$ m/s. The heterogeneous clutter texture parameter is set to $\nu = 25$, and the SCR is set to -2 dB. Fig. 9 provides the SAR image in clutter background and the SAR image of moving targets in the absence of clutter, respectively. The red box in the SAR image under the clutter background represents the real location of the moving target. It can be observed that due to the influence of radial velocity, the Doppler center of the moving targets deviates from zero frequency, resulting in the imaging results being offset in the azimuth direction compared to the placement positions.

Fig. 10(a), (d), and (g), respectively, provides the clutter suppression results for each method, along with the locally magnified results for target T11. It can be seen that since KA-STAP does not perform sample selection, the training samples used to estimate the CCM contain moving targets. As a result, the clutter suppression result exhibits energy cancellation at the moving targets, leading to a decrease in target energy gain. The application of GIP-NHD to KA-STAP has improved this issue to some extent, but there is still a partial loss of energy. The KA-STAP (LE-NHD) scheme based on geometric mean estimation exhibits minimal energy loss and demonstrates the optimal processing performance. Fig. 10(b), (e), and (h) shows the detection results before clustering for each method, while Fig. 10(c), (f), and (i) presents the detection results after clustering. The detection threshold is determined by setting an empirical false alarm probability of $P_{\mathrm{FA}} = 10^{-5}$. Red boxes indicate correct detections, yellow boxes indicate missed detections, and black boxes indicate false alarms. The detection results before and after clustering for each method are shown in Table III. From the detection results, it can be observed that due to the energy loss caused by target cancellation and inaccurate CCM estimation, both the KA-STAP and KA-STAP (NHD) methods result in varying degrees of missed detections after clustering.

TABLE III
DETECTION RESULT IN SIMULATED CLUTTER

Methods	KA-STAP	KA-STAP (NHD)	KA-STAP (LE-NHD)
Missed detections before cluster	0	0	0
False alarms before cluster	7	7	7
Missed detections after cluster	2	1	0
False alarms after cluster	0	0	0

C. Detection Results in Measured Clutter

This section provides the detection results based on the MSTAR-measured clutter background. The depicted clutter scene is a representative heterogeneous environment featuring trees and roads, with eight moving targets arrayed along the road. The SCR of this scene is -4 dB. The 2-D velocity of the moving-target set in this experiment is shown in Table IV. Fig. 11(a) and (b) provides the SAR image in the clutter

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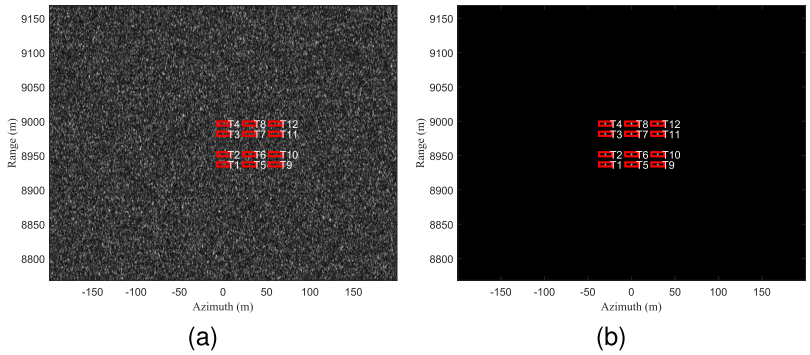


Fig. 9. Scene image in the simulated clutter and target image. (a) Scene image. (b) Target image.

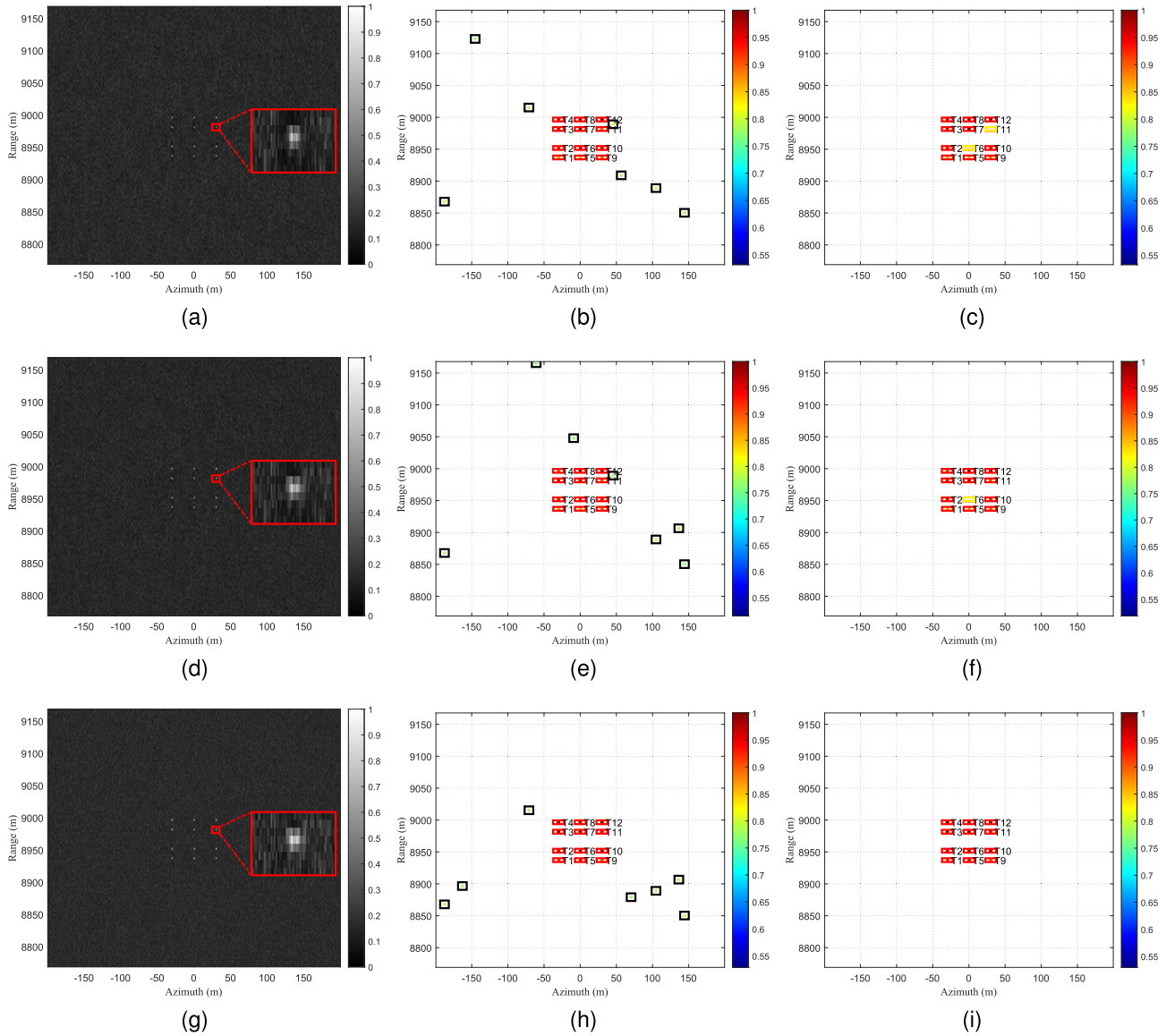


Fig. 10. Detection results under simulated clutter. (a) Clutter suppression result of KA-STAP. (b) Detection result before the cluster of KA-STAP. (c) Detection result after a cluster of KA-STAP. (d) Clutter suppression result of KA-STAP (NHD). (e) Detection result before the cluster of KA-STAP (NHD). (f) Detection result after a cluster of KA-STAP (NHD). (g) Clutter suppression result of KA-STAP (LE-NHD). (h) Detection result before the cluster of KA-STAP (LE-NHD). (i) Detection result after cluster of KA-STAP (LE-NHD).

background and the SAR image of moving targets in the absence of clutter, respectively. For this clutter scene, the heterogeneous clutter statistical model described in (11) and the homogeneous clutter statistical model are used for fitting, and the Kolmogorov–Smirnov (K–S) test is used to verify the fitting accuracy. The pdf of the homogeneous clutter power theoretical distribution is [43]

$$f_T(t,n) = \frac{n^n}{\Gamma(n)} t^{n-1} e^{-\mathrm{nt}} \tag{39}$$

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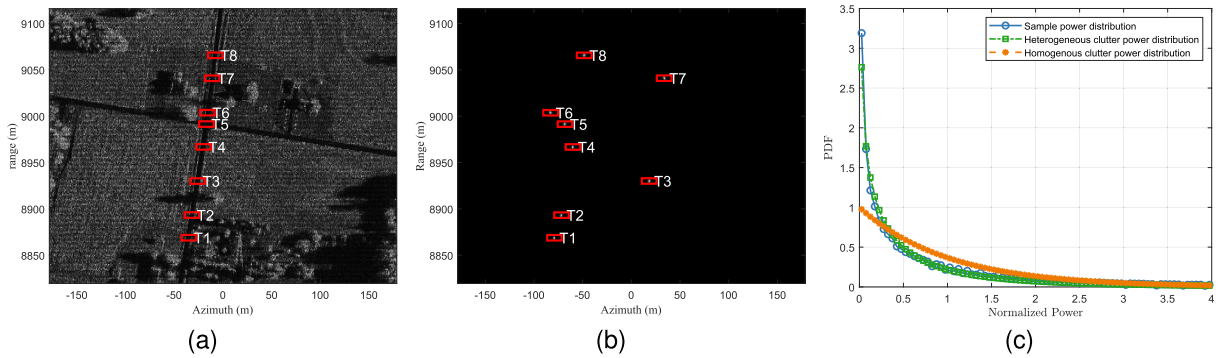


Fig. 11. Scene image in the measured clutter and target image. (a) Scene image. (b) Target image. (c) Statistical distribution.

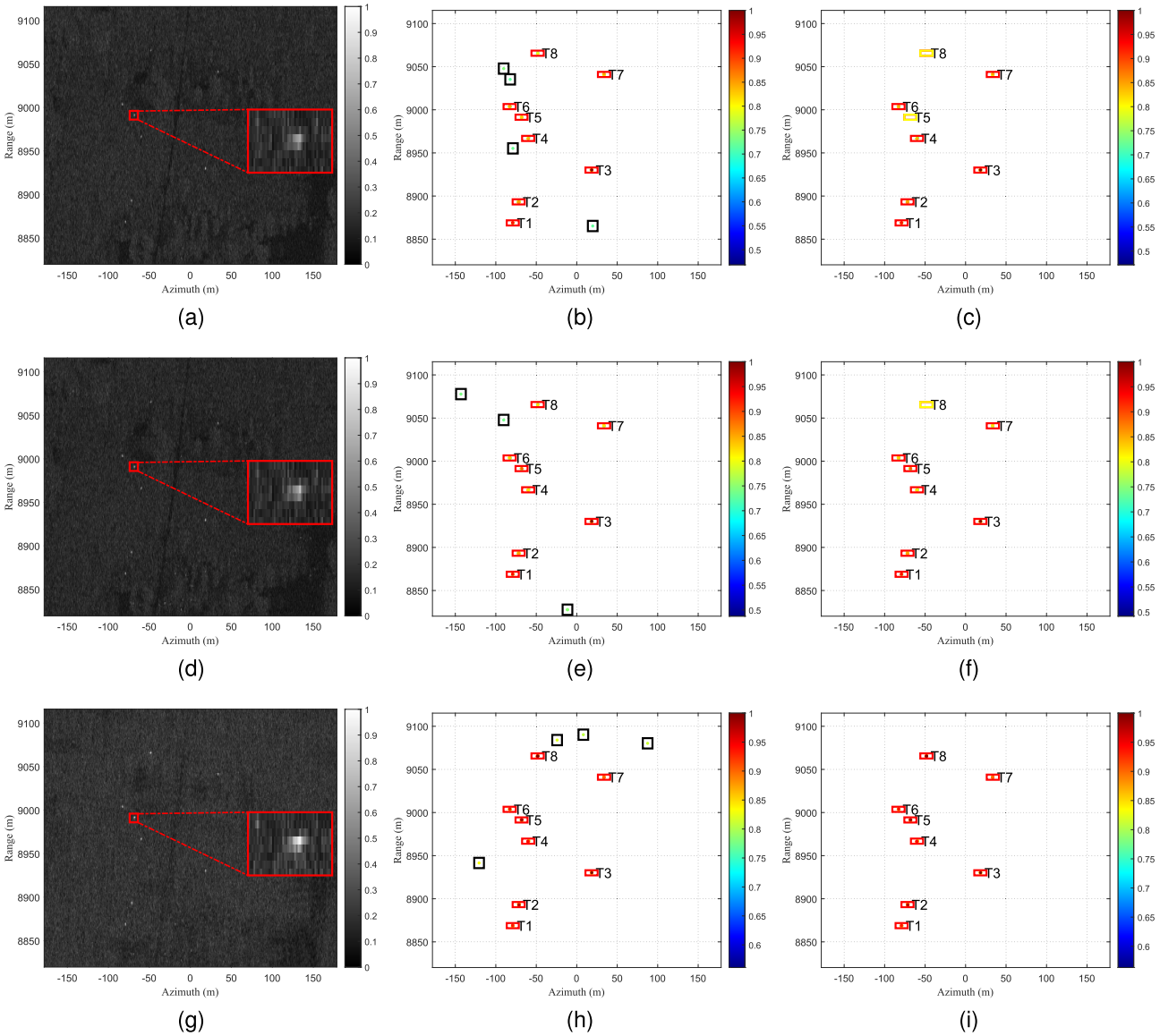


Fig. 12. Detection results under measured clutter. (a) Clutter suppression result of KA-STAP. (b) Detection result before the cluster of KA-STAP. (c) Detection result after a cluster of KA-STAP. (d) Clutter suppression result of KA-STAP (NHD). (e) Detection result before the cluster of KA-STAP (NHD). (f) Detection result after a cluster of KA-STAP (NHD). (g) Clutter suppression result of KA-STAP (LE-NHD). (h) Detection result before the cluster of KA-STAP (LE-NHD). (i) Detection result after cluster of KA-STAP (LE-NHD).

The K–S test statistic is expressed as

$$d_{KS} = \max_{-\infty<x_i<+\infty} \left| \hat{F}_T(x_i) - F_T(x_i) \right| \tag{40}$$

where $\hat{F}_X(x)$ represents the sample distribution function and $F_X(x)$ represents a certain theoretical distribution function. For different distribution functions, the smaller d_{KS}, the higher the degree of fit. The fitting results are shown in Fig. 11(c), the homogeneous clutter model deviates significantly from the actual statistical distribution of the clutter, whereas the heterogeneous clutter model shows a smaller difference from

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TABLE IV
PARAMETERS OF MOVING TARGETS

Target	$x(m)$	$R_B(m)$	$v_x(m/s)$	$v_r(m/s)$
T1	-35	8869	2.05	1.48
T2	-32	8893	2.33	1.33
T3	-26	8930	2.81	-1.48
T4	-20	8967	1.70	1.33
T5	-17	8991	-3.21	1.71
T6	-16	9004	2.43	2.23
T7	-11	9041	1.00	-1.48
T8	-8	9066	1.51	1.34

TABLE V
DETECTION RESULT IN MSTAR CLUTTER

Methods	KA-STAP	KA-STAP (NHD)	KA-STAP (LE-NHD)
Missed detections before cluster	0	0	0
False alarms before cluster	4	3	4
Missed detections after cluster	2	1	0
False alarms after cluster	0	0	0

the actual statistical distribution. The clutter texture parameter is $\nu = 1.9875$, and the K–S test values of the two are $d_{\mathrm{KS-heter}} = 0.021459$ and $d_{\mathrm{KS-homo}} = 0.177941$, respectively. This indicates that the statistical model of heterogeneous clutter described in (11) has better fitting accuracy and is valid and reasonable.

Fig. 12 shows the clutter suppression and moving-target detection results for different methods under the MSTAR-measured clutter background. It can be observed that the KA-STAP method still suffers from target energy loss, whereas the KA-STAP (NHD) and KA-STAP (LE-NHD) methods sequentially mitigate the issue of target energy loss. It is also noticed that both KA-STAP and KA-STAP (NHD) fail to completely suppress discrete clutter such as road edges and trees due to the absence of geometric mean in CCM estimation, while KA-STAP (LE-NHD) has a better suppression effect on such clutter. Notably, when examining the suppression of discrete clutter at the bottom of the scene, it is found that the KA-STAP (NHD) method, compared to the KA-STAP method, actually leaves more residual discrete clutter. This is because, after selecting the training samples, the KA-STAP (NHD) method, with fewer available samples, may have an estimated covariance precision that is sometimes even lower than that of the KA-STAP method. However, the KA-STAP (LE-NHD) method, based on geometric mean estimation, maintains a higher estimation precision even under sample-limited conditions, which ensures the robustness of its clutter suppression performance. Table V summarizes the detection results before and after clustering for each method, indicating that the proposed KA-STAP (LE-NHD) method possesses superior detection performance.

V. CONCLUSION

This article focused on the issue of clutter suppression and moving-target detection for airborne multichannel SAR in heterogeneous clutter environments and proposed a multichannel SAR moving-target detection framework based on the HPD manifold in heterogeneous clutter. First, the representation of airborne multichannel SAR echo data on the HPD manifold is established. Then, using the constructed HPD manifold, the problem of CCM estimation is transformed into a geometric mean estimation problem on the HPD manifold, while the GIP criterion is jointly employed for training sample selection, mitigating the negative impact of training samples contaminated by moving targets. Finally, this approach is applied to the KA-STAP for heterogeneous clutter suppression. Experiments show that using HPD manifold geometric mean estimation for CCM with sample selection outperforms traditional methods in Euclidean space. By evaluating different geometric metrics methods from three aspects: output SCNR, the probability of moving-target samples being selected, and computational complexity, the log-Euclidean metric is selected as the geometric metric criterion for multichannel SAR moving-target detection, considering both performance and computational complexity. The experiments demonstrate that the proposed method can robustly and efficiently estimate the CCM under sample-limited conditions, and its clutter suppression performance and detection capability are verified on both simulated and measured clutter backgrounds.

APPENDIX A
SOLUTION OF THE OPTIMIZATION PROBLEM DEFINED BY (21)

Since the squared Euclidian norm $\|\cdot\|^2$ leads a strictly convex function in the set of complex matrices $R \in \mathbb{C}^{n\times n}$. Notice that when $\|\hat{\boldsymbol{R}}_k\|^2 \to \infty$ and $\|z_k z_k^{\mathrm{H}} - \hat{\boldsymbol{R}}_k\| \to \infty$, there exists a unique optimal solution to problem $\mathcal{P}_{\mathrm{opt}}$. The problem defined by (21) is equivalent to

$$\mathcal{P}'_{\mathrm{opt}}: \begin{cases} \min\limits_{\hat{\boldsymbol{R}}_k} & -2\mathrm{tr}\left\{\hat{\boldsymbol{R}}_k z_k z_k^{\mathrm{H}}\right\} + \mathrm{tr}\left\{\hat{\boldsymbol{R}}_k{}^2\right\} \\ \text{s.t.} & \hat{R}_k \succeq \boldsymbol{I}\ . \end{cases} \tag{41}$$

To facilitate derivation, we first give the following lemma, the proof of which can be found in [51].

Lemma 1: Let $\boldsymbol{A} \succeq 0$ and $\boldsymbol{B} \succeq 0$ be two semi-positive definite matrices, $\lambda_1(\boldsymbol{A}) \geq \lambda_2(\boldsymbol{A}) \geq \cdots \geq \lambda_N(\boldsymbol{A})$ and $\lambda_1(\boldsymbol{B}) \geq \lambda_2(\boldsymbol{B}) \geq \cdots \geq \lambda_N(\boldsymbol{B})$ be the eigenvalues of $\boldsymbol{A}$ and $\boldsymbol{B}$ in decreasing order, respectively. There exists a permutation π of the integers $1, 2, \ldots, N$ such that

$$\mathrm{tr}(\boldsymbol{AB}) = \sum_{i=1}^{N} \lambda_i(\boldsymbol{A})\lambda_{\pi(i)}(\boldsymbol{B}). \tag{42}$$

The rank 1 matrix $z_k z_k^{\mathrm{H}}$ has a unique nonzero eigenvalue $\|z_k\|^2$. Let the eigenvalues of $\hat{\boldsymbol{R}}_k$ be $\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_N)$. From Lemma 1, we know that $\mathrm{tr}\left(\hat{\boldsymbol{R}}_k{}^2\right) = \sum_{i=1}^{N} \lambda_i^2$ and $\mathrm{tr}\left(\hat{\boldsymbol{R}}_k z_k z_k^{\boldsymbol{H}}\right) = -2\|z_k\|\lambda_1$. The optimal solution to the optimization problem defined by (41) is given by $\boldsymbol{U}_k diag(\lambda)\boldsymbol{U}_k{}^{\mathrm{H}}$, where $\boldsymbol{U}_k$ is a unitary matrix of the eigenvectors of $z_k z_k^{\mathrm{H}}$ with the first eigenvector corresponding to the eigenvalue $\|z_k\|^2$. The optimization problems can be rewritten as

$$\mathcal{P}''_{\mathrm{opt}}: \begin{cases} \min\limits_{\lambda} & -2\|z_k\|^2\lambda_1 + \sum\limits_{i=1}^{N} \lambda_i^2 \\ \text{s.t.} & \lambda_i \geq 1, i = 1, \ldots, N. \end{cases} \tag{43}$$

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Consequently, we have $\lambda_i = 1, i = 2, \ldots, N$, and λ_1 is the optimal solution to

$$\mathcal{P}'''_{opt-\lambda_1}: \begin{cases} \min\limits_{\lambda_1} & -2\|z_k\|^2\lambda_1 + \lambda_1^2 \\ \text{s.t.} & \lambda_1 \geq 1. \end{cases} \tag{44}$$

Hence, we have $\lambda_1 = \max\left(1, \|z_k\|^2\right)$ and $\lambda = \left(\max\left(1, \|z_k\|^2\right), 1, \ldots, 1\right)$.

Appendix B
Derivation of Typical Geometric Means on HPD Manifold

Riemannian Mean

To facilitate the derivation, utilizing the relationship between the matrix F-norm and eigenvalues, the Riemannian Distance can be rewritten as follows:

$$\begin{aligned} d_R(\boldsymbol{R}_1, \boldsymbol{R}_2) &= \left\| \log\left(\boldsymbol{R}_1^{-\frac{1}{2}} \boldsymbol{R}_2 \boldsymbol{R}_1^{-\frac{1}{2}} \right) \right\|_F \\ &= \left\| \log\left(\boldsymbol{R}_1^{-1} \boldsymbol{R}_2 \right) \right\|_F \\ &= \left(\frac{1}{K} \sum_{k=1}^{K} \log^2\left(\lambda_i^2\right) \right)^{1/2} \end{aligned} \tag{45}$$

where $\lambda_i, i = 1, \ldots, N$ are the eigenvalues of $\boldsymbol{R}_1^{-1}\boldsymbol{R}_2$ and $\boldsymbol{R}_1^{-(1/2)}\boldsymbol{R}_2\boldsymbol{R}_1^{-(1/2)}$ is similar to $\boldsymbol{R}_1^{-1}\boldsymbol{R}_2$, thus they share the same eigenvalues. Let $F(\boldsymbol{R}) = (1/K)\sum_{k=1}^{K} \| \log(\boldsymbol{R}^{-1}\hat{\boldsymbol{R}}_k\|^2$, we find the gradient and set it to zero

$$\nabla F(\boldsymbol{R}) = \frac{1}{K} \sum_{k=1}^{K} 2 \log(\hat{\boldsymbol{R}}_k^{-1}\boldsymbol{R})\boldsymbol{R}^{-1} = 0. \tag{46}$$

The Riemannian mean satisfies the following relationship:

$$\sum_{k=1}^{K} \log(\hat{\boldsymbol{R}}_k^{-1}\boldsymbol{R}) = 0. \tag{47}$$

To facilitate the derivation, we first present the following lemma [46].

Lemma 2: For any invertible matrix $\boldsymbol{A}$ and matrix $\boldsymbol{B}$, the following relationship holds:

$$\boldsymbol{A} \log \boldsymbol{B}\boldsymbol{A}^{-1} = \log\left(\boldsymbol{A}\boldsymbol{B}\boldsymbol{A}^{-1}\right). \tag{48}$$

Lemma 3: If $\overline{\boldsymbol{R}}$ is the Riemannian mean of a set of HPD matrices $\{\hat{\boldsymbol{R}}_i\}_{1\leq i\leq K}$, then for any generalized linear group $\boldsymbol{C} \in GL(N, \mathbb{C})$, $\boldsymbol{C}\overline{\boldsymbol{R}}\boldsymbol{C}^{\mathrm{T}}$ is the Riemannian mean of the set of HPD matrices $\left\{\boldsymbol{C}\hat{\boldsymbol{R}}_i\boldsymbol{C}^{\mathrm{T}}\right\}_{1\leq i\leq K}$.

By utilizing Lemma 3, one can simultaneously multiply both sides of each $\hat{\boldsymbol{R}}_k$ by $\hat{\boldsymbol{R}}_1{}^{-1/2}$ and still derive (47). Therefore, without loss of generality, it can be assumed that $\hat{\boldsymbol{R}}_1 = I$, and subsequently

$$\log(\boldsymbol{R}) = -\sum_{k=2}^{K} \log(\hat{\boldsymbol{R}}_k^{-1}\boldsymbol{R}). \tag{49}$$

By left-multiplying (49) by $\boldsymbol{R}^{1/2}$ and right-multiplying by $\boldsymbol{R}^{-1/2}$ and utilizing Lemma 2 for simplification, we obtain

$$\log(\boldsymbol{R}) = -\sum_{k=2}^{K} \log\left(\boldsymbol{R}^{1/2}\hat{\boldsymbol{R}}_k^{-1}\boldsymbol{R}^{1/2}\right). \tag{50}$$

Let $\boldsymbol{S} = \log(\boldsymbol{R})$, then (50) becomes

$$\boldsymbol{S} = -\sum_{k=2}^{K} \log(\exp(\boldsymbol{S}/2)\hat{\boldsymbol{R}}_k^{-1}\exp(\boldsymbol{S}/2)). \tag{51}$$

Using the fixed-point method, (51) can be solved iteratively based on the following expression:

$$\begin{aligned} \boldsymbol{S}_0 &= \frac{1}{K}\sum_{k=1}^{K} \log(\hat{\boldsymbol{R}}_k) \\ \boldsymbol{S}_{t+1} &= \beta\boldsymbol{S}_t + (\beta-1)\sum_{k=2}^{K} \log\left(\exp(\boldsymbol{S}_t/2)\hat{\boldsymbol{R}}_k^{-1}\exp(\boldsymbol{S}_t/2)\right) \end{aligned} \tag{52}$$

where $\beta \in (0, 1)$, hence,

$$\begin{aligned} \log(\boldsymbol{R}_{t+1}) &= \beta\log(\boldsymbol{R}_t) + (\beta-1)\sum_{k=2}^{K} \log\left(\boldsymbol{R}_t^{1/2}\hat{\boldsymbol{R}}_k^{-1}\boldsymbol{R}_t^{1/2}\right) \\ &= (\beta-1)\sum_{k=2}^{K} \log\left(\boldsymbol{R}_t^{1/2}\hat{\boldsymbol{R}}_k^{-1}\boldsymbol{R}_t^{1/2}\right) + \log(\boldsymbol{R}_t). \end{aligned} \tag{53}$$

By taking the exponential of both sides and setting $\epsilon_t = 1-\beta$, we obtain

$$\overline{\boldsymbol{R}}_{t+1} = \overline{\boldsymbol{R}}_t^{1/2} \exp\left[\varepsilon_t \sum_{k=1}^{K} \log\left(\overline{\boldsymbol{R}}_t^{-1/2}\hat{\boldsymbol{R}}_k\overline{\boldsymbol{R}}_t^{-1/2}\right)\right]\overline{\boldsymbol{R}}_t^{1/2}. \tag{54}$$

A. Log-Euclidean Mean

Let $F(\boldsymbol{R}) = (1/K)\sum_{k=1}^{K} \| \log(\boldsymbol{R}) - \log\left(\hat{\boldsymbol{R}}_k\right)\|^2$, find the gradient and set it to zero

$$\nabla F(\boldsymbol{R}) = \frac{1}{K}\sum_{k=1}^{K} 2(\log(\boldsymbol{R}) - \log(\hat{\boldsymbol{R}}_k))\boldsymbol{R}^{-1} = 0 \tag{55}$$

$$\sum_{k=1}^{K} (\log(\boldsymbol{R}) - \log(\hat{\boldsymbol{R}}_k)) = 0 \tag{56}$$

Hence, we obtain

$$\hat{\boldsymbol{R}}_{\mathrm{LE}} = \exp\left(\frac{1}{K}\sum_{k=1}^{K} \log\ \hat{\boldsymbol{R}}_k\right). \tag{57}$$

B. JS Divergence Mean

Let $F(\boldsymbol{R}) = (1/K)\sum_{k=1}^{K} \log\left|(\boldsymbol{R} + \hat{\boldsymbol{R}}_k/2)\right| - \log(|\boldsymbol{R}\hat{\boldsymbol{R}}_k|)^{1/2}$, find the gradient and set it to zero

$$\begin{aligned} \nabla F(\boldsymbol{R}) &= \frac{1}{2K}\sum_{k=1}^{K}\left(\left(\frac{(\boldsymbol{R} + \hat{\boldsymbol{R}}_k)}{2}\right)^{-1} - \boldsymbol{R}^{-1}\right) \\ &= 0. \end{aligned} \tag{58}$$

Simplifying the above equation, we obtain

$$\boldsymbol{R}^{-1} = \frac{1}{K}\sum_{k=1}^{K}\left(\frac{\boldsymbol{R} + \hat{\boldsymbol{R}}_k}{2}\right)^{-1}. \tag{59}$$

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Taking the inverse of both sides of the equation

$$\boldsymbol{R} = \left(\frac{1}{K} \sum_{k=1}^{K} \left(\frac{\boldsymbol{R} + \hat{\boldsymbol{R}}_k}{2} \right)^{-1} \right)^{-1}. \tag{60}$$

Using the Weiszfeld algorithm, the iterative solution for the geometric mean of the JS divergence can be obtained

$$\overline{\boldsymbol{R}}_{t+1} = \left(\frac{1}{K} \sum_{k=1}^{K} \left(\frac{\overline{\boldsymbol{R}}_t + \hat{\boldsymbol{R}}_k}{2} \right)^{-1} \right)^{-1}. \tag{61}$$

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